Signal-Adaptive GraphRAG for Traditional Chinese Bible Question Answering: End-to-End Architecture, Knowledge-Graph Quality Remediation, and Rank Fusion

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***Abstract*—Domain-specific question answering (QA) with large language models (LLMs) faces high API cost, data-privacy constraints, and strict factuality requirements. We present the complete design, deployment, and four-round empirical evaluation of a graph-based retrieval-augmented generation (GraphRAG) system for Traditional Chinese Bible QA, followed by tests of graph retrieval on the rebuilt system. The legacy system of Rounds 0–3 converted 66 books of the Revised Chinese Union Version from PDF into a pericope-centric hierarchical corpus by layout-geometry analysis, applied token-aware hierarchical chunking, and built a knowledge graph with grounded dual-track entity extraction (9,120 entities of six types; 173,896 mentions), closed-schema grounded relation extraction (37 relation types), and a 250,418-edge cross-reference network that integrated the Treasury of Scripture Knowledge. Content, vectors, and graph were stored in PostgreSQL, Qdrant, and Neo4j, and served online by a signal-driven six-route retrieval architecture with eleven strategies, cross-encoder reranking, and a linear rank-fusion layer $s_{\text{fused}} = (1-\alpha)\, s_{\text{rr}} + \alpha\, w$; an October 2026 rebuild retired Neo4j and reduced the knowledge layer to a name layer (2,926 names) and a curated registry of 31 events. On a 100-question benchmark spanning five question types we report a three-round evidence chain: (i) signal-adaptive retrieval beats a semantic-only baseline by $+14$–15 percentage points hit rate with either answer model; (ii) a knowledge-graph completeness overhaul (entity anchor coverage $83.8\% \to 98.2\%$, cross-references $\times 273$) yields *zero* retrieval gain on the head-entity benchmark — a negative result we diagnose, question by question, to a reranker-monopolized final ranking, topical cross-reference noise, and benchmark head saturation; and (iii) a one-day rank-fusion remediation recovers and surpasses all prior results on the 100 questions it was developed on (hit rate 0.97, event-narrative hit rate 1.00, $+0.16$ over the semantic baseline), with an ablation showing a complementary structure between discrete data repairs and continuous score fusion. A fourth round (Round 3) re-evaluates on 500 questions (the original 100 plus 400 in 19 diagnostic families) with verse-level metrics, a repaired faithfulness judge, a repaired intent classifier (silently inactive in all earlier local-model runs), and a no-graph ablation: routing alone accounts for $+0.13$ of the replicated $+0.16$ head margin, and the graph's $+0.03$ is not distinguishable from zero (95% CI upper bound $+0.08$). On the 399 new questions no graph gain is detected (verse recall@5 -0.009 and answer coverage -0.014, each with upper bound $+0.001$, from one run per configuration); the direction is consistently negative, but the anchor-coverage@5 deficit of -0.012 has a Holm-adjusted $p = 0.107$, and harm survives multiplicity adjustment only in first-gold-rank win:loss counts (12:35, sign test $p = 0.001$, replicated in a rerun; a statistic chosen after inspecting the results), a direction we trace to wrongly anchored graph candidates passing the ranking layer; faithfulness is ≈ 0.98 throughout. On the rebuilt system, whose knowledge layer has no entity-identity or relation layer, we tested graph operations over a name–passage graph — chiefly personalized PageRank (PPR), used directly and as name hints for an LLM sub-query writer — against dense retrieval that reads the same number of passages (k-aligned) and, for the sub-query arms, against non-graph twins of the same pipeline, on multi-hop questions written by an LLM and not reviewed by humans. No comparison is confirmatory, and none detected a gain outside the development set: dense-seeded PPR, selected on that set, where it won 11 question clusters and lost none at $k = 10$, split 2:2 in a pre-registered check on a second, partially exposed set and was stopped (this is not evidence of absence); PPR name hints changed hop coverage by -0.017 against their non-graph twin (upper bound $+0.023$); and apparent gains of graph-hinted sub-queries were attributable to LLM query rewriting. We relate this null to reranker selection, missing identity and relation layers, and little graph-unique headroom, and we stopped graph-retrieval work on this layer pending a new graph layer and a new pre-registration. We conclude that, in the legacy rounds, knowledge-graph construction quality, the retrieval ranking mechanism, and benchmark design *jointly* determined observable benefit and that optimization order should follow the bottleneck layer rather than the upstream-first order of the data flow; and, across both systems, that a layer's gains must be validated by ablation on questions its repairs never saw and, for a graph, against both k-aligned dense retrieval and a non-graph twin.**

***Index Terms*—Evaluation methodology, GraphRAG, knowledge graph construction, negative results, question answering, rank fusion, retrieval-augmented generation, Traditional Chinese.**

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Manuscript drafted July 7, 2026; revised October 2, 2026 to add the September 2026 expanded-benchmark evaluation (Round 3). Corpus and graph statistics were re-verified against the legacy deployment, then live, on July 6, 2026; Round 3 statistics are recomputed from the archived run outputs. Statements contradicted by an October 2026 data audit were corrected on October 8, 2026 (Appendix C); no experimental number was changed. Revised again on October 18, 2026 to add graph-retrieval experiments run after Round 3, mostly on the rebuilt system (Section XII), to add multiplicity-adjusted readings of Round 3, and to mark the retired graph-centric architecture as legacy; no number reported for Rounds 0–3 was changed.

I. Introduction

LARGE language models (LLMs) excel at open-domain question answering, but deploying them for a specific knowledge domain exposes three persistent obstacles. First, commercial API pricing places a recurring cost on every query, which is prohibitive for resource-constrained organizations such as churches, seminaries, and non-profit publishers.

Second, sensitive corpora and user questions often cannot leave the premises, ruling out cloud-only inference. Third, and most specific to the religious domain, factual grounding is non-negotiable: recent measurements report that even state-of-the-art LLMs score lowest on faith-related dimensions of a human-flourishing alignment benchmark [1], reinforcing a legitimate concern in the Christian community about whether LLM answers are actually grounded in Scripture rather than in the model's parametric memory.

Retrieval-augmented generation (RAG) [2] mitigates factual errors by conditioning generation on retrieved evidence, and graph-based RAG (GraphRAG) variants [3], [4] additionally exploit entity and relation structure for multi-hop and aggregation queries. However, most published GraphRAG systems are evaluated as end-to-end black boxes: when a knowledge graph (KG) improvement does or does not move the final metric, the community rarely learns *which layer* — extraction, storage, retrieval, ranking, generation, or the benchmark itself — gated the outcome. This paper documents a deployed system in which exactly that question was forced into the open, four times on the legacy system by concrete engineering incidents, and once more, under pre-registered stopping rules, on the rebuilt system.

We designed, built, deployed, and iteratively optimized a GraphRAG system for question answering over the Traditional Chinese Bible (Revised Chinese Union Version; RCUV). The system evaluated in Rounds 0–3, which we call the legacy system, converted 66 books from PDF into a pericope-centric hierarchical corpus, constructed a knowledge graph with grounded entity and relation extraction, indexed three complementary databases (PostgreSQL, Qdrant, Neo4j), and served queries through a signal-driven six-route retrieval architecture terminated by a cross-encoder reranker and a linear rank-fusion layer. Along the way we performed a quantified audit of KG construction quality, executed a completeness overhaul, observed a rigorously verified *negative result* — the overhaul produced zero retrieval improvement on our benchmark — diagnosed the mechanism question by question, and repaired the actual bottlenecks (the ranking and vocabulary layers) in one day of engineering, surpassing all previous results on the same benchmark. A September 2026 re-evaluation on an expanded 500-question benchmark, with a no-graph control, then replicated that head result while showing that most of it belongs to signal routing rather than to the graph. An October 2026 audit of the data pipeline later found defects in the corpus conversion and the retrieval stack that qualify these results, which we report as run; the system has since been rebuilt and no longer uses Neo4j or any graph traversal (Appendix C). The rebuild reduced the knowledge layer to a name layer and a curated event registry, with no entity-identity or relation layer. On it we tested graph retrieval — personalized PageRank (PPR) over a name–passage graph, used directly, as a source of name hints for an LLM sub-query writer, and as extra candidates in iterative retrieval — against k-aligned dense retrieval and, where an LLM rewrote the query, against non-graph twins, under pre-registered stopping rules; no gain was detected outside the development set, and we closed graph retrieval on the current knowledge layer (Section XII).

This paper makes eight contributions. Contributions 1–7 report on the legacy system of Rounds 0–3, with pointers to the rebuilt system where its evidence bears on them; contribution 8 concerns the rebuilt system.

1) **A complete pipeline from raw PDF to GraphRAG for Traditional Chinese biblical text, and an audit of its graph** (Sections III and IV): layout-geometry PDF parsing without OCR or LLMs, pericope-centric hierarchical chunking with the embedding model's own tokenizer, grounded dual-track entity extraction (six types, 9,120 entities, 173,896 mentions), closed-schema grounded relation extraction (37 relation types with evidence-span verification), and a three-source cross-reference network of 250,418 edges integrating the Treasury of Scripture Knowledge (TSK) in its OpenBible.info edition (CC BY) [5]. This is the legacy pipeline. A later audit showed that its completeness metrics did not imply correctness (the strict precision of its 15,926 semantic edges was about 0.23–0.28), and its graph was retired rather than rebuilt (Appendix C).
2) **A signal-adaptive six-route retrieval architecture over a triple-database backend, terminated by a rank-fusion layer** (Section V): in the legacy system, six boolean query signals selected among six routes plus a fallback; each route ran a weighted combination of eleven retrieval strategies; candidates were deduplicated, reranked by a cross-encoder, fused by $s_{\text{fused}} = (1 - \alpha)\, s_{\text{rr}} + \alpha\, w$, and finally adjusted by a small set of "user-literal" pins. Its measured value came mainly from routing, not graph traversal: Round 3 attributes $+0.13$ of the $+0.16$ head margin to routing (Section XI), and on the rebuilt system the pipeline's answer-coverage gain over semantic-only retrieval ($+0.032$, descriptive; Appendix C) is concentrated in the 179 questions with explicit scripture references ($+0.092$), that is, in citation resolution; on the other 321 questions no gain was detected (-0.002, upper bound $+0.017$).
3) **A quantified causal chain from chunking granularity to KG quality** (Section VIII): we show that the damage to KG quality was caused not by chunk-size parameters but by the *silent inheritance* of an embedding-oriented granularity structure by the extraction, import, and retrieval layers (mechanisms M1–M6), producing a 55.9% silent mention loss and 1,474 retrieval-blind entities. The finding is aligned with, and locally re-validates, published evidence on extraction granularity [3], document-level relation extraction [6], and coreference-resolved KG construction [7]–[9].
4) **A verified negative result with a mechanism-level diagnosis** (Section IX): a KG completeness overhaul that raised entity anchor coverage from 83.8% to 98.2% and multiplied cross-references by 273 produced *no* improvement — and locally a regression — on a 100-question head-entity benchmark. Question-by-question forensics attribute this to (a) topical cross-reference votes acting as noise on event-narrative questions, (b) a final ranking monopolized by the cross-encoder's

surface-form score, and (c) benchmark head saturation. This result complements benchmark-level observations that KG defects propagate non-linearly to answers [10] and that GraphRAG gains are frequently over-read due to evaluation bias [11].

5) **A rank-fusion remediation with an interpretable ablation** (Section X): TSK de-noising by vote-weighted edge declassing, a curated head-event backfill that re-enabled entity-vector retrieval, and the linear fusion layer together lift hit rate to 0.97 (event-narrative questions to 1.00), surpassing the pre-overhaul state and widening the margin over the semantic-only baseline to $+0.16$ on the 100 development questions; Round 3 re-attributes most of that margin to routing (Section XI). The α ablation exposes a complementary structure — discrete data repairs saturate hit rate, continuous fusion recovers tail anchors — and a mechanism-level finding that discrete "pin" patches flip from net-positive to net-negative once continuous fusion exists.
6) **An expanded-benchmark re-evaluation with a no-graph ablation** (Section XI): on 500 questions (the original 100 plus 400 in 19 diagnostic families), with verse-level retrieval metrics and a repaired faithfulness judge, the head result replicates (hit rate 0.97, $+0.16$) but decomposes into $+0.13$ from signal routing and a $+0.03$ from the graph that is not distinguishable from zero (95% CI upper bound $+0.08$), the latter concentrated in hand-curated head events. On the 399 new questions no graph gain is detected (verse recall@5 -0.009 and answer coverage -0.014, each with upper bound $+0.001$, from one run per configuration). The direction is consistently negative, but after Holm adjustment over eight retrieval tests the anchor-coverage@5 deficit (-0.012) has $p = 0.107$; the harm survives only as first-gold-rank win:loss counts, under a Holm adjustment applied to the sign tests of the same eight cells (12:35, sign test $p = 0.001$, replicated at 13:32, $p = 0.007$, in a retrieval-only rerun), a statistic chosen after inspecting the results. We trace this direction to wrongly anchored graph candidates passing a ranking layer that cannot tell them from right ones. The round also documents two instrument failures of general relevance: a faithfulness judge blind to the generator's citation headers (0.912 rising to 0.987 on unchanged answers once repaired), and an intent classifier silently disabled by a reasoning model's hidden tokens.
7) **A methodological conclusion for GraphRAG practice** (Section XIII): KG construction quality, the retrieval ranking mechanism, and benchmark design jointly determine the observable benefit of any single-layer investment. In the legacy deployment, one day of ranking-layer repair outperformed what a week-scale re-extraction was projected to deliver on the same benchmark, an in-sample comparison; three incidents show graph signal gated by retrieval-side mechanisms, a fourth shows wrong graph evidence passing through them, and the expanded benchmark shows that a head benchmark can misattribute gains between layers. On the rebuilt system, measured against k-aligned dense retrieval and, where an LLM rewrote the query, non-graph twins on targeted multi-hop questions, graph signal was not detectable outside the development set (contribution 8). We argue that evaluation design is a *prerequisite* for — not an afterthought of — further construction-side optimization, and that a layer's contribution must be established by ablation on questions its repairs were not developed on.
8) **A negative result for graph retrieval on the rebuilt system, with upper bounds, an attribution protocol, and three failure mechanisms** (Section XII). Each graph arm was compared with dense retrieval reading the same number of passages and, where it used an LLM rewrite, also with a non-graph twin of the same pipeline, on multi-hop sets written by an LLM and not reviewed by humans; only PPR or traversals of two or more hops counted as graph operations (a rule fixed before the October 18 run and applied retrospectively to the earlier runs). No result is confirmatory, and none detected a gain outside the development set: PPR-ranked name hints for an LLM sub-query writer changed hop coverage by -0.017 against their non-graph twin (upper bound $+0.023$), and dense-seeded PPR, selected on the development set (cluster win:loss 11:0 at $k = 10$; upper bound $+0.054$ at the production $k = 5$), split 2:2 in a pre-registered check on a second, partially exposed set and was stopped (not evidence of absence). We describe three mechanisms consistent with this null — reranker selection, no identity or relation layer, and little graph-unique headroom — and the conditions for reopening the line (Sections XII-C–XII-D).

The remainder of the paper follows the pipeline order: corpus and preprocessing (Section III), the legacy knowledge-graph construction (Section IV) and online retrieval and generation (Section V), the evaluation framework (Section VI), the quality audit and the four experiment rounds (Sections VII–XI: Round 0, the P0 round as Round 1, the rank-fusion round as Round 2, and the expanded-benchmark Round 3), graph retrieval on the rebuilt system (Section XII), discussion (Section XIII), limitations and future work (Sections XIV–XV), and conclusion (Section XVI). Appendix A maps each pipeline stage to the literature it draws on.

II. Related Work

A. Retrieval-Augmented Generation and Hybrid Retrieval

RAG [2] couples a parametric generator with non-parametric retrieval; surveys [12] catalogue its now-extensive design space. Dense passage retrieval [13] and lexical scoring [14] remain the two complementary retrieval substrates, commonly combined by reciprocal rank fusion (RRF) [15]. Our system uses BGE-M3 [16] for dense embeddings (and its companion cross-encoder, BGE-reranker-v2-m3, for reranking); the legacy system also indexed BM25 sparse vectors over Chinese-specific CKIP tokenization [17] for RRF dense–sparse hybrid search, but that path never executed (Section V-E), so every run of Rounds 0–3 is dense-only. Distinct from

standard hybrid search, the final ranking in our system linearly fuses the cross-encoder score with a *strategy-prior* weight that encodes which retrieval strategy proposed the candidate — in the legacy system, a channel by which graph evidence survived into the final top-k (Section V-F). Calibrated fusion of heterogeneous graph and vector scores has recently been studied in multi-hop QA [18]; our fusion is deliberately simpler (a single linear interpolation with one global α) and is evaluated by ablation.

B. Graph-Based RAG

GraphRAG [3] popularized LLM-built entity graphs with hierarchical community summaries for query-focused summarization; subsequent systems explore lighter indexing and different retrieval operators, e.g., LightRAG's dual-level keyword retrieval [19], personalized-PageRank retrieval over passage-linked graphs [20], attributed-community hierarchical retrieval [21], and recursive abstractive summary trees (RAPTOR) [22]. Event-centric variants build explicit event nodes and temporal edges [23], [24]. Benchmark studies ask *when* graphs actually help RAG [25]. Our legacy architecture differed from these retrieval-operator designs in that graph traversal was one family among the eleven strategies selected *per query* by boolean signals (route-level adaptivity), and in that the paper's empirical focus is not a new operator but the interaction between graph asset quality, ranking, and benchmark — an interaction that, to our knowledge, has not been documented at incident-level granularity for a deployed GraphRAG system. On the rebuilt system we do evaluate an operator of the PPR family [20], seeded by dense retrieval over a name–passage graph, though not as a reimplementation of that system: each graph arm is held to k-aligned dense retrieval and, where an LLM rewrites the query, to a non-graph twin, which bears on the question of when graphs help [25] for a graph without entity identity (Section XII).

C. LLM-Based Knowledge Graph Construction

LLM-driven KG construction is surveyed in [26], which contrasts schema-based and schema-free extraction paradigms; our relation extractor is schema-based with a closed 37-type ontology. On construction quality, GraphRAG's own ablation shows chunk granularity directly controls extraction density (600- vs. 2,400-token chunks yield nearly twice the entity references) [3]; DocRED quantifies that at least 40.7% of relational facts require cross-sentence evidence and 17.6% require coreference reasoning [6]; and "lost-in-the-middle" effects bound how much context can be packed into a single extraction call [27]. Coreference-resolved construction reduces node duplication by 28–45% in the CORE-KG / LINK-KG line of work [7]–[9], with the ablation in [8] cleanly separating coreference (deduplication) from structured prompting (noise suppression). Post-extraction correction is an alternative to re-extraction: ontology-grounded repair [28], canonicalization after open extraction [29], KG denoising that *shrinks* graphs yet improves downstream RAG [30], retrieval-augmented construction that re-retrieves cross-chunk context for each pre-entity [31], restoring the source context of isolated triples [32], and GNN-guided cross-chunk graph augmentation [33]. Atomic-fact decomposition with dual-time modeling [34] shows finer construction units increase fact coverage. Our audit (Section VIII) maps each of our locally measured defects onto this literature, and the remediation roadmap we proposed adopted its recipes. Beyond the P0 overhaul, that roadmap was not executed: re-extraction (P1) was deferred (Section IX), and the legacy graph was retired in the October 2026 rebuild (Appendix C).

D. Evaluating RAG and GraphRAG

We evaluate generation with RAGAS [35] plus a custom LLM-judged answer-coverage metric, and retrieval with deterministic ground-truth metrics. Two recent results shape our methodology. BRINK benchmarks how KG-based RAG breaks under incomplete knowledge and finds that answer quality degrades non-linearly with KG defects, with systems partly relying on parametric memory rather than structural reasoning [10]. An unbiased evaluation framework for GraphRAG shows that after removing position and verbosity bias from LLM-judged comparisons, reported win rates can drop by tens of points (e.g., LightRAG $66.70\% \rightarrow 39.06\%$) [11]. Our own evaluation encountered a concrete instance of judge bias — the RAGAS answer-relevancy "noncommittal" rule zeroes honest refusals, deflating the score of a cautious commercial model — and a larger one in the judge's inputs: a faithfulness judge that never saw the generator's citation headers scored correct citations as unsupported (Section VI-B). Chunking-strategy sensitivity for *retrieval* (not construction) has been studied systematically in [36]; we cite it only for its retrieval-side scope, since no peer-reviewed quantification exists for overlap effects on *extraction*, and we rely on DocRED [6] as indirect evidence there.

III. Corpus and Data Preprocessing

The offline pipeline (Fig. 1) turns 66 books of the RCUV from PDF into (i) a hierarchical structured corpus, (ii) a multi-granular embedding queue, and (iii) the inputs of knowledge-graph construction. All build steps after the PDF conversion run on the host from a dedicated `scripts/` environment; every intermediate artifact is materialized as JSONL before database import, so each stage is independently inspectable and re-runnable.

A. Source Corpus

The source is 66 per-book PDFs (one per biblical book). Their copyright pages label the text the Chinese Union Version (CUV) with new punctuation, but the label is wrong: the proper-name spellings (e.g., for Reuben, Cyrus, Caesar, and Darius) identify the text as the Revised Chinese Union Version (RCUV, *Shangti* edition). The CUV is the de facto standard Protestant translation in the Traditional Chinese world, and its revision, the RCUV, keeps much of its early-20th-century literary vocabulary, which creates a systematic lexical gap between modern user phrasing and corpus surface forms — a gap that recurs throughout this paper (e.g., the modern phrase "the Last Supper" never occurs in the RCUV text, which narrates a "Passover meal").

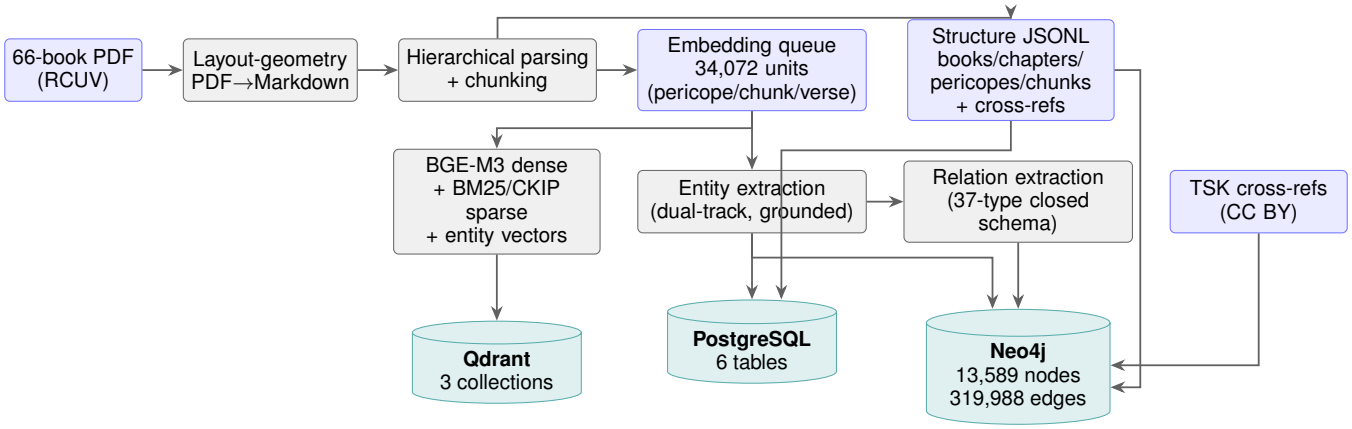


Fig. 1. Offline build pipeline. Every arrow is a materialized JSONL hand-off; the three databases played complementary roles in Rounds 0–3 (Section V-A); database counts show the legacy deployment's post-remediation state (July 6, 2026; Neo4j was retired in October 2026, Appendix C). The embedding queue doubles as the extraction feed — the structural root cause analyzed in Section VIII (mechanism M1).

B. PDF-to-Markdown by Layout Geometry

Conversion (`convert_bible_pdf.py`, PyMuPDF) uses *pure layout geometry* over character-level spans — no OCR and no LLM — so the step is free and, for a fixed PyMuPDF version, deterministic:

- font size ≤ 8.5 pt: running headers/footers, dropped;
- font size ≥ 16 pt: book title or chapter number;
- font size $= 12$ pt with $x \geq 85$: pericope (section) heading;
- font size ≤ 9.5 pt and numeric: verse number;
- verse number at $x < 75$ with line right-edge < 300: poetry (line breaks preserved); otherwise prose (lines joined);
- bold `N:N` tokens: footnotes, aggregated per chapter.

Limitation of the legacy conversion. An October 2026 audit (Appendix C) found this converter lossy. Footnotes that cross a page break, headings or a "Selah" inside a verse, indented lines, and glyphs set beyond the right page edge (which the default extraction clips) removed text from 268 verses (4,446 characters); 67 further verses lost only their "Selah"; all 116 psalm superscriptions (1,490 characters) were dropped; and 61 genuine headings were lost while seven verse fragments became headings. The step ran outside the build pipeline, with no PyMuPDF version pinned in any of the repository's environments, and its Markdown output carries 39 lines of undocumented manual edits that merged 17 textual-variant readings into the text, creating 10 *ghost verses* that the PDF does not print and altering 10 others. All corpus counts in this paper, and every index, graph, and experiment of Rounds 0–3, were built from this legacy corpus; we report them as run.

The resulting Markdown uses `#` for book, `##` for chapter, `###` for pericope headings, and bold verse numbers. Under each pericope heading, the RCUV's printed parallel-passage annotations (e.g., "(Mark 1:9–11; Luke 3:21–22)" under the baptism pericope) are retained — they become the first source of the cross-reference network (Section IV-C).

C. Pericope-Centric Hierarchical Structure

A *pericope* is the established interpretive unit of biblical scholarship: a self-contained narrative episode, discourse, or psalm. The RCUV's editorial pericope headings give us this segmentation for free, and the system adopts the pericope as the core unit for embedding, entity anchoring, and cross-referencing, for three reasons:

1) **Semantic completeness.** A story is never split down the middle; retrieval returns interpretable wholes.
2) **Embedding fit.** The median pericope is 388 characters (P90 = 931, max = 3,544), well inside the effective input range of the BGE-M3 embedder [16].
3) **Explainability.** Returning "Isaac takes a wife (Gen. 24)" is more interpretable to end users than a verse-range span.

Parsing (`process_bible.py` with `markdown_parser.py`) yields the hierarchy Book → Chapter → Pericope with counts 66 / 1,189 / 2,779, plus per-pericope verse spans. A regex-based reference parser with a Chinese book-abbreviation table resolves the parallel-passage annotations; every reference is *upgraded to pericope level* (never verse level), keeping the cross-reference network aligned with the retrieval unit. Hierarchical IDs are colon-delimited (`gen` → `gen:1` → `gen:1:0` for the first pericope of Genesis 1), a scheme reused verbatim across all three databases.

D. Hierarchical Chunking

Long pericopes are split for embedding by a token-aware hierarchical chunker whose tokenizer is *BGE-M3's own*, i.e., the chunking layer is explicitly designed for the embedding model [16]:

- pericopes with ≤ 768 tokens (93.9%) are kept whole;
- longer pericopes are split on *verse boundaries* targeting 512 tokens per chunk, with a 1-verse overlap between adjacent chunks;
- a trailing chunk under 128 tokens and ≤ 3 verses is merged back into its predecessor;

TABLE I
CORPUS AND CHUNKING STATISTICS (LEGACY CONVERSION OF THE RCUV PDFS, USED IN ROUNDS 0–3)

Item	Value
Books	66
Chapters	1,189
Pericopes	2,779
– split into chunks	169 (6.1%)
Chunks (from split pericopes)	431
Corpus size (characters)	1,329,046
– inside split pericopes	243,773 (18.3%)
Pericope length, median / P90 / max (chars)	388 / 931 / 3,544
Split-pericope length, median / max (chars)	1,353 / 3,544
Chunk length, median / max (chars)	644 / 796
Chunker: split threshold / target / min (tokens)	768 / 512 / 128
Chunker: overlap	1 verse
Tokenizer	BGE-M3

TABLE II
ENTITY TYPES (LEGACY GRAPH COUNTS, JULY 6, 2026; RETIRED OCTOBER 2026)

Type	Examples	Count
Person	Abraham, Moses, Jesus	2,419
Object	Ark of the Covenant, tabernacle	2,200
Event	the Exodus, the Resurrection	1,714
Place	Jerusalem, Egypt	1,299
Theme	redemption, grace, faith	987
Group	Israelites, Pharisees	505
Total		9,124

The deployed counts differ slightly from the raw extraction output (9,120; Section IV-A) because of import-time merging and the post-audit curation of Sections IX–X (16 generic-noun events deleted, 18 curated events added, one entity retyped); the pre-remediation deployed count, used by the audit of Section VIII, was 9,122.

- chunk IDs extend the parent ID (`gen:1:0:0`), preserving the hierarchy.

Only 169 of 2,779 pericopes (6.1%) are split, producing 431 chunks; the split pericopes concentrate in Old Testament narrative books (Leviticus 13, Judges 12, 1 Samuel 12, Genesis 11, Numbers 11, Joshua 10, Deuteronomy 9, 2 Samuel 9) and account for 18.3% of the corpus text (243,773 of 1,329,046 characters); their median length is 1,353 characters (max 3,544), against a chunk median of 644 (max 796). Each retrieval unit's embedded text carries a locating prefix — "*Book, Chapter N, pericope title (verses X–Y):*" — so vectors encode provenance context. Table I summarizes the corpus. Its tokenizer row refers to the build side; in Rounds 0–3 the online query encoder ran an older tokenizer-library version that encoded queries inconsistently with the build side (Section VI-C).

E. Multi-Granularity Embedding Queue

Retrieval needs different granularities for different question shapes, so the pipeline emits an embedding queue that is the union of three levels, 34,072 units in total:

- **pericope level** (2,610): every pericope that was *not* split;
- **chunk level** (431): split pericopes enter only as chunks;
- **verse level** (31,031 verse labels, 10 of them ghost verses): every verse individually, for fine-grained exact retrieval (printed verse ranges such as "1–2" are kept as single units).

This queue is an artifact designed *for retrieval*. Critically for what follows, the early entity-extraction pipeline consumed this same queue as its input feed. That single wiring decision — extraction feed = embedding queue — silently propagated an embedding-oriented granularity into the knowledge graph and became the structural root cause (M1) of the defect chain quantified in Section VIII.

IV. KNOWLEDGE GRAPH CONSTRUCTION

This section describes the legacy knowledge graph of Rounds 0–3, which was retired in October 2026 (Appendix C); the reduced knowledge layer of the rebuilt system is described in Section XII-A. The legacy graph comprised three subnetworks — the corpus hierarchy, the entity network, and the cross-reference network — built by the four pipelines described below. Throughout construction we enforced a single anti-hallucination design rule, *grounding*: LLMs never invent entities, relation types, or evidence; they only *classify* candidates that rule-based stages propose, and every LLM output must quote an evidence span that is verified post hoc to be a literal substring of the source text, otherwise the output is demoted or discarded (the run of record also contains edges without verified evidence; they are listed at the end of Section IV-B). This is the schema-based end of the paradigm spectrum surveyed in [26], and the structured-prompting choice follows the noise-suppression evidence of [7], [8].

A. Grounded Dual-Track Entity Extraction

Entity extraction targets six types (Table II) and runs two tracks in parallel: a named-entity-recognition (NER) track and a grounded classification track.

1) NER Track (Person / Place / Group): CKIP Transformers [17] performs Chinese word segmentation and named entity recognition (BERT-base models), with the type mapping PERSON→Person, GPE/LOC→Place, ORG/NORP→Group. Results are merged with a hand-built biblical entity dictionary using dictionary-first, longest-match precedence. Mention positions are located by literal string matching (`text.find()`) — a design whose consequences (no coreference; substring false hits) are quantified in Section VIII.

2) Grounded Track (Event / Object / Theme): Four phases, rule-first with LLM escalation:

1) **Pericope-title mining.** Titles are typed by suffix and keyword heuristics (altar/vessel/ark suffixes → Object; redemption/grace keywords → Theme; residual default → Event). A 24-word generic-title stoplist ("day(s)", "firstborn", ...) intercepts generic nouns before the Event default — a guard added after the noise audit of Section VIII.
2) **POS candidate mining.** CKIP part-of-speech tagging proposes nouns (Na/Nb/Nv/Nc) with corpus frequency

TABLE III
CLOSED RELATION ONTOLOGY (37 TYPES)

Type pair (count)	Relations
Person×Person (13)	FATHER_OF, MOTHER_OF, SON_OF, DAUGHTER_OF, SIBLING_OF, SPOUSE_OF, ANCESTOR_OF, DESCENDANT_OF, TEACHER_OF, DISCIPLE_OF, ENEMY_OF, ALLY_OF, SUCCEEDED_BY
Person×Place (7)	BORN_IN, DIED_IN, RULED, VISITED, EXILED_TO, RETURNED_FROM, BUILT
Person×Object (4)	POSSESSED, RECEIVED, GAVE, DESTROYED
Person×Event (3)	PARTICIPATED_IN, INITIATED, VICTIM_OF
Person×Group (3)	MEMBER_OF, LEADER_OF, FOUNDED
Place×Place (2)	LOCATED_IN, NEAR
Event×Place (1)	OCCURRED_IN
Event×Event (2)	PRECEDED_BY, CAUSED
Group×Place (2)	ORIGINATED_FROM, SETTLED_IN

Each type defines domain/range, direction, inverse type, regex trigger signals, few-shot examples, and stage-specific confidence priors.

≥ 3; adjacent nouns are compounded (e.g., "burnt" + "offering").

3) **Rule classifier.** Exact lexicons and verb-component features assign types; candidates with confidence ≥ 0.8 are finalized without any LLM call.
4) **LLM classifier (grounded).** Remaining candidates are classified by a small local model (Gemma 3 4B [37]) under two hard constraints: it may only label *provided* candidates (never add new ones), and it must return an evidence quotation that post-hoc verification confirms is a substring of the grounding text; failures are demoted.

An `EntityNormalizer` merges canonical names and aliases within each type and assigns `entity_id` = `{type}:{pinyin}` (toneless `lazy_pinyin` — a homophone-collision risk noted in Section XIV). The extraction output is 9,120 entities and 173,896 mentions; mentions carry the granularity of their source unit (pericope, chunk, or verse), inherited from the embedding queue (Section III-E). Known upstream limitations, all revisited in the audit: a stopword list removes 33 single-character words including "God", "Spirit", and "Lord"; there is no coreference resolution; entity descriptions are 100-character text snippets; and pinyin IDs can collide across homophones.

B. Grounded Relation Extraction with a Closed 37-Type Ontology

Entity–entity relation edges are extracted by a five-stage pipeline against a closed ontology of 37 relation types (Table III), each declaring domain/range types, direction, inverse, textual trigger signals, few-shot examples, and per-stage confidence priors. Closing the schema prevents the LLM from inventing relation names, and typing the domain/range prunes the candidate space before any model call.

1) **Pair mining.** Entity pairs co-occurring in the same pericope are enumerated, kept only if some ontology type admits their type signature, and capped at 80 pairs per pericope.
2) **Rule classifier.** Per-type regex trigger signals fire when the two entities are within 25 characters; matches with confidence ≥ 0.85 are finalized.
3) **Domain priors.** 71 curated golden genealogy relations bypass the LLM entirely (64 of them enter the graph as distinct prior-stage edges; the remainder coincide with earlier-stage matches).
4) **Grounded LLM classification.** A local Gemma 4 31B model [38] — the relation stage was built after the entity stage and uses a larger model; the entity classifier's 4B scale is itself flagged in the audit — must select exactly one type from the pair's type-legal candidate set *or* answer NONE, under a JSON grammar constraint; the returned `evidence_span` must pass substring verification against the source pericope (doubly checked post hoc), otherwise the pair is rejected. This addresses the classic failure mode of triples divorced from their source context [32].
5) **Inverse materialization.** Symmetric/inverse types (FATHER_OF $\leftrightarrow$ SON_OF) are materialized bidirectionally at confidence $\times 0.9$.

The full run (10–20 hours offline, checkpointed and resumable) yields 6,958 imported relations (772 rule, 64 prior, 5,370 LLM, 752 inverse) — and, importantly, 77,953 *unclassified* candidate pairs (LLM answered NONE) retained on disk with their source-pericope provenance. This "rejected pile" later became the raw material of the P0 event-layer rescue (Section IX). A companion description generator (Gemma 4 E4B) targets the $\sim$4,223 Person/Place/Group entities with empty descriptions, generating ≤ 80-character summaries grounded in the titles of mentioning pericopes; entities without pericope anchors cannot be described this way — which is why the audit of Section VIII still finds 46% of NER-type entities description-less.

Relation edges have `confidence`, `evidence_span`, `source_pericope_id`, and `extraction_phase` fields — the property that later allowed co-occurrence backfill edges (confidence 0.35) to coexist distinguishably with LLM-extracted edges. The fields record provenance, not correctness, and are not uniformly filled: the October 2026 audit (Appendix C) found no source pericope on the 64 prior edges and no evidence on the 9,060 co-occurrence backfill edges of Section IX, and two independent samples put the strict precision of all 15,926 semantic edges (Table IV) at about 0.23–0.28 (lenient 0.34–0.45). They are therefore candidate relations, not verified facts — one part of failure mechanism (B) of Section XII-C.

C. Cross-Reference Network

Cross-references ("chain references") connect pericopes across books — prophecy and fulfillment, parallel gospel accounts, quotations. The network merges three sources of decreasing editorial cost:

1) **Printed parallel-passage annotations** parsed from the RCUV Markdown: 774 edges (synoptic parallels and psalm titles);

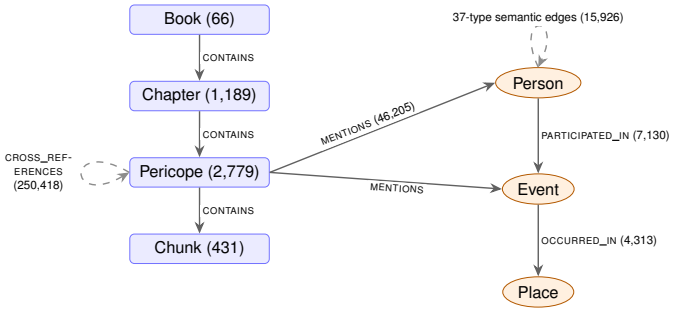


Fig. 2. Knowledge-graph schema: corpus hierarchy (left), entity network (right), the pericope-to-pericope cross-reference network (774 markdown + 142 curated + TSK edges), and sequential NEXT / NEXT_BOOK edges (omitted for clarity). Verse-level nodes deliberately do not exist — the design trade-off whose downstream cost is quantified in Section VIII.

2) **A curated list of famous New-Testament-to-Old-Testament (NT→OT) citations**: 142 edges typed as *quotation* or *allusion*;
3) **The Treasury of Scripture Knowledge (TSK)** [5], a cross-reference compendium, used in its OpenBible.info edition with community vote counts (licensed CC BY, not public domain): 344,799 raw rows are filtered (1,166 negative-vote rows, 9,811 self-loops) and mapped verse→pericope through a reverse index built from the embedding queue (covering 31,102 verse positions, with printed verse ranges expanded — a figure that includes the 10 ghost verses of Section III and equals the KJV verse total only by coincidence; the PDF has 31,092 verse positions); only 7 rows fail to map, yielding 250,358 unique pericope pairs.

After idempotent merge the legacy graph held **250,418** CROSS_REFERENCES edges (TSK is not loaded in the rebuilt system; Appendix C). Every TSK edge stores its `votes`; manual edges are assigned a sentinel vote of 999 (highest trust). This vote-based provenance is what later made *selective de-noising* possible without deleting data (Section X) — the same "repair by re-weighting, not removal" philosophy advocated for KG denoising in [30].

D. Graph Schema and Storage

Figure 2 shows the schema; Table IV the scale of the legacy graph as deployed. Salient design points:

- **Dual labels.** Hierarchy nodes carry `(:Pericope:Bible)`, entity nodes `(:Person:Entity)`: the generic label supports whole-structure queries, the specific label exact matches.
- **No verse nodes.** Verse-level IDs (`gen:1:0:v:3`) exist only in Qdrant and PostgreSQL. This was a deliberate economy at design time — and the direct precondition of the silent mention-loss defect (Section VIII, M2).
- **Entity properties.** `canonical_name`, `aliases` (a native list), `description`, `mention_count`.
- **Provenance-first edges.** Semantic edges keep confidence and evidence fields; cross-reference edges keep votes and source; backfilled edges are flagged — all imports were idempotent MERGE operations, so while the legacy graph was deployed it could be rebuilt or rolled back deterministically. Neo4j and the legacy data were deleted in the October 2026 rebuild; the legacy graph can now be restored only from the pre-rebuild backup (Appendix B).

TABLE IV
LEGACY KNOWLEDGE-GRAPH SCALE (NEO4J, JULY 6, 2026; RETIRED OCTOBER 2026)

Item	Count
Nodes, total	13,589
structure (Book/Chapter/Pericope/Chunk)	4,465
entities (six types)	9,124
Relationships, total	319,988
MENTIONS	46,205
CROSS_REFERENCES	250,418
semantic edges (37 types)	15,926
PARTICIPATED_IN / OCCURRED_IN	7,130 / 4,313
SON_OF / FATHER_OF / POSSESSED	659 / 647 / 551
structural: CONTAINS / NEXT / NEXT_BOOK	4,399 / 2,975 / 65

E. Vector Indexes and Database Import

Three vector families are produced:

- **Dense**: BGE-M3 [16], 1,024-dimensional, normalized, for all 34,072 queue units;
- **Sparse**: BM25 [14] (k_1=1.5, b=0.75) over CKIP-segmented [17] text, with the query-side IDF vocabulary exported for the online service (indexed, but never used for online scoring; Section V-E);
- **Entity vectors**: each entity is embedded from the recipe "*name (aliases). description. commonly appears in: pericope titles*" truncated to 200 characters, upserted idempotently (UUID5 keys) into a dedicated collection (9,122 points).

In the legacy system, imports populated PostgreSQL (6 tables; the authoritative store), Qdrant (`bible_embeddings`, dense-only; `bible_embeddings_hybrid` with named dense+sparse vectors; `bible_entities`), and Neo4j (structure, mentions, semantic edges, cross-references). The import layer was where the audit later found the single most damaging defect of the whole pipeline — verse-level mentions silently dropped because their target nodes do not exist (Section VIII) — and where honesty counters were subsequently installed.

V. ONLINE RETRIEVAL ARCHITECTURE

The online service evaluated in Rounds 0–3 was a FastAPI application exposing `POST /api/v1/query`. Figure 3 shows its query pipeline; this section walks through that legacy architecture stage by stage. The service rebuilt in October 2026 no longer queries Neo4j and has no graph traversal: its routes are served by SQL and dense strategies only, the reranker and rank fusion are kept without graph candidates, and the knowledge layer enters retrieval only through a curated event lookup lane (its names serve only as routing vocabulary; Appendix C). The graph-retrieval experiments on the rebuilt system are reported in Section XII.

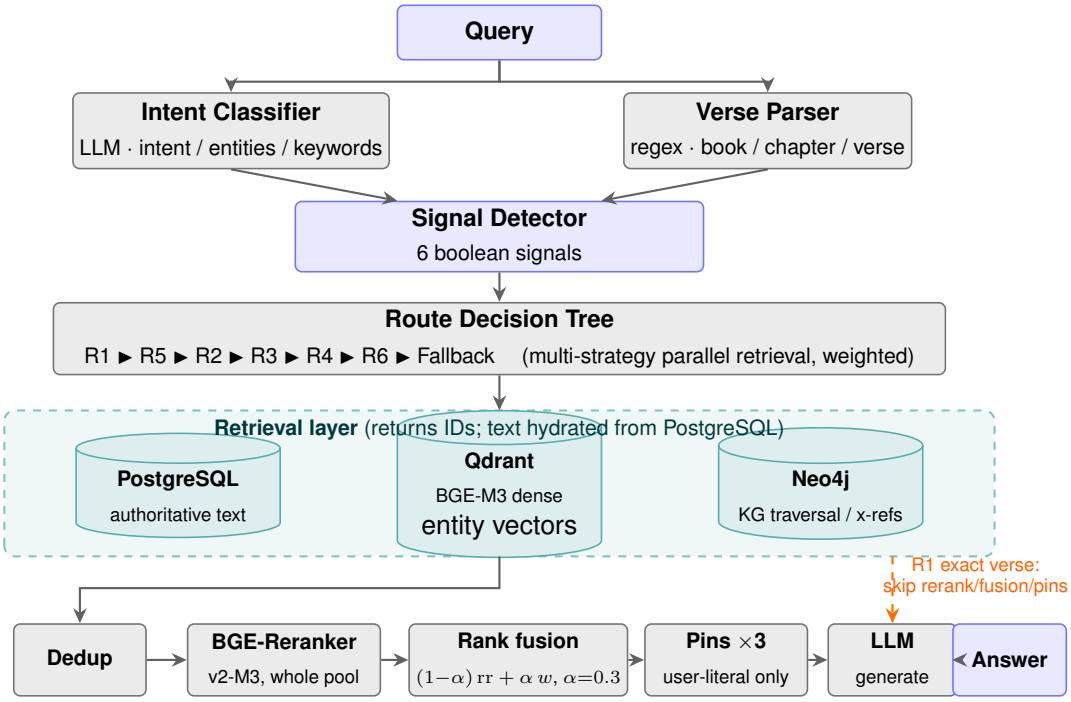


Fig. 3. Legacy online query pipeline (Rounds 0–3; Neo4j and the graph strategies were removed in the October 2026 rebuild). Compared with the May 2026 deployment, the post-retrieval chain gained the rank-fusion stage, and the pin set was re-audited: only "user-literal" pins survive (Section V-F).

A. Triple-Database Roles and Content Hydration

In Rounds 0–3 the three stores played strictly complementary roles:

- **PostgreSQL** was the authoritative source of truth: full text, hierarchy, entity mentions. *Every* retrieval strategy returns only IDs; final passage text is always hydrated by a single PostgreSQL lookup (`get_content_by_id()`), which resolves 3-segment IDs to pericopes, 4-segment IDs to chunks, and verse IDs to their parent pericope. This guarantees that no matter which index proposed a candidate, the generator always sees identical, authoritative text.
- **Qdrant** was the semantic entry point: dense and entity-vector search (the configured hybrid mode never executed; Section V-E).
- **Neo4j** was the relational entry point: hierarchy traversal, entity networks, and cross-reference expansion. It was removed in the October 2026 rebuild.

Strategy weights are *priors* on the reliability of the proposing strategy, not similarity scores; deduplication keeps the maximum weight per ID and weights are only ever upgraded. Every retriever is wrapped in per-strategy error isolation: a failing strategy is recorded in a `strategy_errors` field of the response and never aborts the query.

B. Query Pipeline

A query passes through seven steps: (1) verse-reference detection (regex); (2) LLM intent classification returning {`intent, entities, keywords`}; (3) signal detection and route selection; (4) parallel multi-strategy retrieval for the selected route; (5) deduplication, whole-pool reranking, rank fusion, and pins; (6) context assembly from the top-5 passages; (7) constrained answer generation. Five request-level switches matter for evaluation: `semantic_only` bypasses steps 1–3 (the ablation baseline); `use_graph` gated all Neo4j strategies per request in Rounds 0–3 (A/B without container restarts; in the rebuilt service it gates only the curated event lookup lane); `fusion_alpha` overrides α (Section X-C); `retrieval_only` skips step 7 (the fast evaluation loop of Section VI-C); and `include_context` returns, per source, the context block the generator saw, for judging. Responses expose observability fields — per-source strategy, raw and fused scores, route used, strategies used, and errors — which later made question-by-question forensics possible without re-instrumentation.

C. Intent Classification and Signal Detection

- **Verse parser**: regex over canonical and abbreviated Chinese book names (longest-match, length-descending) resolves "Romans 3:23"-style and "Genesis chapter 1"-style references.
- **Intent classifier**: the generation LLM returns JSON with `intent` $\in$ {`verse_lookup`, `topic`, `person`, `event`, `cross_reference`} plus entity names and keywords; a detected verse reference forces `verse_lookup`. In the Gemma configurations of Rounds 0–2 this call silently returned nothing — the model's hidden reasoning tokens exhausted a 256-token output budget and the parser fell back to `topic` with empty lists — so those rounds were routed by the verse parser and the dictionaries alone; the budget was raised to 4,096 tokens before Round 3 (Section XI).
- **Dictionaries**: person/place gazetteers are *reused from the build-side extraction dictionary*, matched by longest

Algorithm 1 Signal-driven route selection

Require: signals S (six booleans), intent i

if S.book_chapter_verse **then**
  return R1 (exact verse)
else if S.book_chapter $\wedge$ S.multi_book **then**
  return R5 (cross-book)
else if S.book_chapter **then**
  return R2 (chapter + semantic)
else if S.multi_book $\vee$ $i =$ cross_reference **then**
  return R5 (cross-book)
else if S.multi_person **then**
  return R3 (person graph)
else if S.event_keyword **then**
  return R4 (event graph)
else if S.has_place **then**
  return R6 (place graph)
else
  return Fallback (semantic + book anchor)
end if

substring; event detection uses a curated list of ∼54 event keywords (Sermon on the Mount, division of the kingdom, Paul's conversion, …); book-name matches require $\geq$ 2 characters; LLM-returned entities are re-filtered through the same dictionaries.

Six boolean signals result: *has-book-chapter-verse*, *has-book-chapter*, *multi-book*, *multi-person*, *event-keyword*, *has-place*. There is deliberately *no query rewriting*: the raw user query reaches both the embedder and the reranker verbatim, so the lexical gap between modern phrasing and the CUV-derived vocabulary of the RCUV text must be bridged by graph anchors and the fusion layer — a recurring causal thread in Sections IX–X. The rebuilt-system experiments revisit this design bet (Section XII-C): where graph-derived names were combined with LLM sub-query rewriting, a non-graph twin with the same rewriting reproduced the gain over plain dense retrieval.

D. Six-Route Decision Tree

Different question shapes need different evidence: "What does John 3:16 say?" should be answered by SQL lookup; "the relationship between Moses and Jethro" by person-graph traversal; "how Jeremiah's new-covenant prophecy is applied in Hebrews" by cross-book expansion. A priority-ordered decision tree (Algorithm 1) maps the six signals to six routes plus a fallback; each route launches its strategies in parallel with the priors listed in Table V. This was the design rationale of the legacy routes; on the rebuilt system, a prototype kinship traversal showed no detectable gain over k-aligned dense retrieval on multi-hop questions (exploratory, development set; Section XII-B).

E. Retrieval Strategies

In Rounds 0–3, eleven strategies — three of them the typed graph traversals — shared the ID-only/hydrate-later contract of Section V-A (the Neo4j-backed ones were removed in the rebuild):

- **`semantic`** — BGE-M3 query embedding against the passage collection, top-20. A hybrid mode was designed to prefetch dense and BM25-sparse branches (50 each) and merge them by RRF [15] inside Qdrant, but it never executed: the installed Qdrant client (1.8.2) lacks the `query_points` call it relies on, so every hybrid request fell back to dense-only (541 of 541 logged calls) while still being labeled hybrid. All results of Rounds 0–3 are therefore dense-only.
- **`verse_direct`** — exact SQL lookup of a verse, verse range, or whole chapter.
- **`sql_chapter / sql_supplement`** — direct chapter fetch; and same-chapter completion for chapters already hit by other strategies.
- **`graph_person / graph_event / graph_place`** — entity lookup by canonical name or alias (substring match, ordered by `mention_count`), then MENTIONS traversal to pericopes. Multi-person queries additionally retrieve *co-occurrence* pericopes at weight 0.9. Event

TABLE V
LEGACY ROUTES AND STRATEGY COMPOSITIONS (ROUNDS 0–3; WEIGHTS IN PARENTHESES)

Route	Strategies (strategy prior w)
R1	`verse_direct` (1.0); empty ⇒ fall back to R2
R2	`sql_chapter` (0.9) + `semantic` (0.6)
R3	`graph_person` (0.9) + `semantic` (0.7) + `entity_path` + `cross_ref_expand` + `entity_query` (0.6) + `sql` (0.5) + `book_anchor`
R4	`graph_event` (0.85) + `semantic` (0.7) + `cross_ref_expand` + `entity_query` (0.6) + `sql` (0.5) + `book_anchor`
R5	`cross_reference` (0.85) ‖ `sql_chapter` (0.85) ‖ `graph` (0.75) + `semantic` (0.65) + `entity_query` (0.6) + `sql` (0.4) + `book_anchor`
R6	`graph_place` (0.85) + `semantic` (0.7) + `entity_path` + `cross_ref_expand` + `entity_query` (0.6) + `sql` (0.5) + `book_anchor`
FB	`semantic`/`hybrid` + `book_anchor` (if a book is named)

Names are the legacy deployment's route-configuration keys: `graph` denotes the route's typed entity-graph traversal (`graph_person` / `graph_event` / `graph_place`); `cross_reference` is R5's dedicated cross-reference retrieval, sharing the seed-support+votes machinery of `cross_ref_expand`; `sql` is a low-prior supplementary SQL lookup distinct from `sql_chapter`; `hybrid` is the configured dense+sparse mode of `semantic`, which fell back to dense-only in every reported run (Section V-E). With `use_graph=false`, all Neo4j-backed strategies were gated off. Observed route distribution on the original 100 questions: R1=20, R2=25, R3=21, R4=18, R5=10, R6=4, fallback=2 in the Gemma runs of Rounds 0–1 (R4=19, R6=3 in Round 2, after the event keyword "Gethsemane prayer" was added); R1=20, R2=25, R3=19, R4=21, R5=14, R6=1, fallback=0 in Round 3, with the repaired classifier. On all 500 questions (Round 3, graph run): R1=90, R2=87, R3=74, R4=92, R5=87, R6=9, fallback=61.

anchors are ordered canonically (book/chapter ascending, so the narrative origin surfaces first) and tagged with `keyword_exact`/`anchor_rank` flags for the pin stage. All graph queries remap chunk IDs to their parent pericope *inside Cypher* — a fix that also restored retrieval visibility for 164 chunk-only entities (Section X).

- **`cross_ref_expand`** — $\leq$ 2-hop expansion over CROSS_REFERENCES from seed pericopes, ranked by *seed support* (number of distinct seeds reaching the candidate) then by *votes*; 2-hop only fills unmet quota. Hop weights are provenance-stratified: manual edges 0.75/0.55 (1-/2-hop), TSK edges 0.60/0.50 — deliberately *below* the semantic prior 0.7 after the de-noising episode of Section X; expansion is capped at 10 candidates and seeds are chosen round-robin across strategies to prevent single-strategy monopoly.
- **`entity_path`** — multi-hop reasoning (≤ 2 hops) over the 37-type semantic edges only (structural edges excluded), then MENTIONS to pericopes.
- **`entity_query`** (EQ) — the query embedding matched against *entity* vectors (top-8, cosine ≥ 0.4), then MENTIONS to pericopes, with hub-aware throttling: entities with `mention_count` > 50 contribute at most 3 pericopes, others 5, supplement cap 5. EQ is the designed bridge between modern question vocabulary and the classical corpus; its disable–re-enable history is itself an experimental finding (Sections VII and X).
- **`book_anchor`** — book-filtered semantic search guaranteeing that an explicitly named book contributes seeds.

F. Rank-Fusion Layer

The post-retrieval chain was the architectural core of the legacy system of Rounds 0–3 and the layer where two of the four incidents of Table XXII were resolved and where the fourth arose. The rebuilt service keeps reranking and rank fusion, but no graph candidates reach them. In the legacy system the chain was:

1) **Deduplication**: same-ID candidates keep the maximum strategy prior.
2) **Whole-pool reranking**: BGE-reranker-v2-m3 (the cross-encoder companion of [16]) scores every (query, passage) pair; scores are sigmoid-normalized to $[0, 1]$. The *whole* candidate pool is scored — not a top-k prefix — so fusion can promote candidates the reranker puts arbitrarily deep.
3) **Linear rank fusion**: $s_{\text{fused}} = (1 - \alpha)\, s_{\text{rr}} + \alpha\, w$ with $\alpha = 0.3$, where s_{rr} is the normalized reranker score and w the strategy prior of Table V. α is overridable per request (enabling the sweep of Section X-C); $\alpha = 0$ recovers pure reranker ordering, and a configuration flag restores the full legacy path.
4) **Pins**: a small additive ladder applied after fusion, restricted to "user-literal" triggers (Table VI).

Why fusion was introduced. The cross-encoder is a surface-form matcher. When modern question vocabulary does not occur in the RCUV text, the ground-truth anchor's reranker score can be as low as 0.001 — e.g., "the Last Supper" against a corpus that says "Passover meal" — and no amount of candidate-pool engineering makes it visible under pure reranker ordering. In the legacy system, fusion let a candidate with $s_{\text{rr}} \approx 0.001$ but a graph prior of $w = 0.85$ enter the top-5. The magnitudes are interpretable: at $\alpha = 0.3$ a graph anchor ($w \in [0.85, 0.9]$) beats a TSK neighbor ($w \in [0.5, 0.6]$) unless the latter's reranker advantage exceeds ≈ 0.11. R1 (exact verse) bypasses reranking, fusion, and pins entirely; if the reranker itself fails, ordering falls back to strategy priors. In Round 3 the graph strategies also injected off-target anchors that the reranker, and in some cases the prior, then favored (Section XI-C). On the rebuilt system the same cross-encoder, which scores a passage only against the question or sub-query, underlies the first failure mechanism of graph retrieval, (A) selection: in the per-hop and appended-passage settings, graph-proposed bridge passages reached the candidate pool but rarely survived to the top-k (Section XII-C).

TABLE VI
LEGACY PIN LADDER (ROUNDS 0–3; ADDITIVE BONUSES ON THE FINAL SCORE)

Pin	Bonus	Status	Trigger
chapter	$+0.010$	active	user names "Book, chapter N" [a]
EQ	$+0.005$	legacy only	high-confidence EQ candidate
book-anchor	$+0.004$	active	user names a book
graph uncertainty	$+0.003$	legacy only	reranker top-1 < 0.3
keyword-exact event	$+0.002$	active (fusion)	dictionary event keyword = entity name/alias [b]

[a] Guarantees ≥ 2 passages of the named chapter survive (candidates with $w \geq 0.85$ qualify).
[b] Exact string equality only; hub entities (`mention_count` > 25) excluded; one anchor per event, at most two pins. The bonuses are spaced so pins never override one another.

Pin lifecycle. The EQ pin and the graph-uncertainty pin were designed *before* fusion existed, as discrete rescues for graph evidence that pure reranker ordering was suppressing. Once fusion exposed those same signals continuously, the two pins were re-audited, found to be net-negative (they promoted $s_{\text{rr}} \approx 0.002$ noise to rank 1 on two benchmark questions), and retired from the fusion path — a mechanism-level lesson analyzed in Section XIII. The surviving pins all encode "the user literally named X" constraints, which are orthogonal to score fusion.

G. Answer Generation

The top-5 hydrated passages are formatted as numbered context blocks ("[N] Book, Chapter — pericope title (verse span)"), and the generator operates under a strict system prompt: answer *only* from the provided passages, cite book and chapter for every claim, add no outside knowledge or inference, respond in Traditional Chinese. The LLM provider is pluggable behind a factory (Ollama, Claude, OpenAI, Gemini); the deployed default is a locally served Gemma 4 E4B (8-bit quantized) at temperature 0.1 [38]. If retrieval returns nothing, the system refuses rather than answers. Responses are non-streaming JSON envelopes carrying the answer, per-

source metadata (strategy, raw and fused scores), the route used, and timing — the observability that later enabled per-question forensics.

H. Deployment

In Rounds 0–3 the system shipped as five Docker Compose services — backend (FastAPI), PostgreSQL 15 with pgvector, Qdrant v1.13.2, Neo4j 5.15 Community with APOC, and Ollama with GPU access — with health-checked startup ordering. The backend image was baked (no code volume); the only code-relevant mounts were the read-only BM25 vocabulary (loaded, but unused for scoring; Section V-E) and the HuggingFace model cache. The rebuilt deployment drops the Neo4j service and the BM25 vocabulary mount and serves one versioned build whose contract files are mounted read-only. Build (`scripts/`), serving (`backend/`), and evaluation (`evaluation/`) each pin their own dependency manifest, isolating environment drift. All models — embedder, reranker, extractor, generator, judge — run locally; no external API is required for any pipeline stage, which is the cost/privacy design goal carried over from the system's inception.

VI. Evaluation Framework

A. Benchmark

The benchmark contains 500 Traditional Chinese questions, 100 in each of five categories. Rounds 0–2 (Sections VII–X) used the original 100 questions, 20 per category, with ground truth approved by annotators with biblical expertise; Round 3 (Section XI) uses all 500. The five categories are:

- **VERSE_LOOKUP** — exact reference lookup ("What does Romans 3:23 say?");
- **TOPIC** — thematic aggregation ("What does the Bible teach about forgiveness?");
- **PERSON** — character-centric questions, often multi-person ("What was the relationship between Elijah and Elisha?");
- **EVENT** — event-narrative questions ("What happened at the Last Supper?");
- **GENERAL** — open questions requiring knowledge integration, including cross-book prophecy–fulfillment.

Each item carries `reference` (gold pericope-level provenance), `expected_answer_points` (a checklist of content points), and `reference_answer` (a model answer). The taxonomy deliberately spans exact matching, semantic search, entity-centric retrieval, and cross-book integration, so that route-level behavior is observable per category.

Once Round 2 had saturated the original 100 (Section X), 400 questions were added on July 11, 2026 — after the Round 2 repairs — in 19 diagnostic families of size $n \geq 8$ (Table VII); we call them the *expansion* set and the first 100 the *original* (legacy) set. Each family aims at a failure mode identified in Sections VIII–X. The *longtail_** families ask about books, chapters, persons, and events that head questions never reach, *head_cold* about cold passages of head books, and *parable* about the parables. *Paraphrase* gives a quotation without its reference, which bypasses R1; *title_coref* names a person only by title (the missing-coreference gap) and *disambiguation* asks about homonymous persons; *trajectory* and *multi_chapter* require cross-chapter aggregation (the PERSON_004 family); *split_pericope* targets persons in chunk-split pericopes (mechanism M3); *temporal* asks for event ordering and *surface_form* uses modern labels absent from the RCUV text. Five cross-book multi-anchor families populate the GENERAL_006/008 family of Section XIII. All 400 references were machine-validated against the corpus, every question was content-checked against the corpus text in an LLM-assisted review (49 corrections), and new reference answers use the corpus's own spellings. The original 100 are unchanged, so they remain a comparable subset across all four rounds. Audits in October 2026 found that the 500 questions can reliably detect harm from a graph but not a small gain: 275 of them (55%) give away the answer's location in the question itself (an explicit citation or a single lexical anchor), only one is primarily a multi-hop question, and 15 need two or more hops. The graph tests on the rebuilt system therefore use separate multi-hop question sets, written by an LLM and not reviewed by humans (Section XII-A).

TABLE VII
Benchmark Composition (500 Questions, 100 per Type)

Type	Diagnostic families added in July 2026 (n)
VERSE_LOOKUP	longtail_book (48), head_cold (14), paraphrase (18)
TOPIC	longtail_chapter (35), multi_chapter (25), head_cold (20)
PERSON	longtail_person (30), disambiguation (12), title_coref (12), trajectory (18), split_pericope (8)
EVENT	longtail_event (34), temporal (16), surface_form (16), parable (14)
GENERAL	prophecy_fulfillment (26), nt_quotes_ot (16), typology (12), parallel_account (14), theme_thread (12)

Each type also keeps its 20 original questions (the Round 0–2 benchmark).

B. Metrics

Sixteen metrics (counting the two faithfulness judges separately) fall into four families:

1) **Deterministic retrieval metrics** (no LLM in scoring; recomputable bit-exactly from archived outputs): hit rate, recall@5, precision@5, F1@5, MRR, MAP@5, NDCG@5, computed programmatically against the gold references with each gold pericope or chapter range counted as one unit; and, added in July 2026, *verse recall@5* (the fraction of gold verses retrieved, ranges expanded verse by verse) and *anchor coverage@5* (the fraction of gold chapter-level anchors reached by at least one retrieved passage; not to be confused with the knowledge graph's *entity* anchor coverage of Section IX).
2) **RAGAS generation metrics** [35]: faithfulness, answer relevancy, context recall, and answer correctness, judged by a locally served Gemma 4 26B. Since September 2026 faithfulness is reported by two judges that share one statement decomposition: *zh*, a Traditional Chinese

NLI rubric that also sees the user question, and *strict*, RAGAS's default NLI prompt.

3) **Answer coverage** (ours): an LLM judge scores each expected answer point as covered (1.0), partial (0.5), or missing (0.0); coverage is the mean. Unlike F1-style correctness it is a *pure recall* metric — it rewards completeness without penalizing extra detail, complementing faithfulness as a hallucination check.
4) **Semantic similarity**: embedding cosine between generated and reference answers.

Five measurement pitfalls, encountered concretely in this project, qualify the numbers we report. (i) Per-question judge noise for RAGAS metrics reaches ± 0.4, so 20-question per-category deltas below ≈ 0.05 are treated as noise (Round 3 measures the floor directly, Section XI). (ii) RAGAS answer relevancy applies a "noncommittal" rule that zeroes refusals; an honest "the provided passages do not state this" is scored 0, which we found systematically deflates cautious models (a commercial model's relevancy rose above the local model's once refusal-affected items were excluded). (iii) Position and verbosity bias make naive LLM-judged A/B comparisons overstate differences [11]. (iv) Unit-level recall is inflated by ranges: a gold "Matthew 5–7" (111 verses) is one unit, so retrieving any one of its verses scores recall 1.0. On the Round 3 graph run, unit-level recall@5 is 0.835 but verse recall@5 0.749 (0.893 vs. 0.688 on the original 100); the hit rates and recall@5 values of Rounds 0–2 are unit-level — valid as "some gold passage was retrieved," optimistic about breadth. (v) The faithfulness judge originally received passage text without the generator's citation headers ("[1] John, chapter 3 — *title* (verse 16)"), so the citations the generator is instructed to write were judged unsupported, and short, citation-dense answers suffered most: exact-verse questions, with the best retrieval, scored the lowest faithfulness in every Round 0–2 run. Re-judging the 500 answers of a July 15, 2026 run with context blocks rebuilt in the generator's exact format raised faithfulness from 0.912 to 0.987 (zh) and 0.981 (strict) without changing a single answer. Every faithfulness value in Rounds 0–2 predates this repair and is not comparable with Round 3. Retrieval conclusions in this paper therefore rest on the deterministic family wherever possible.

Following a July 2026 audit of the instruments, answer coverage is the primary answer-side metric and faithfulness a gate — since the September repair, strict faithfulness ≥ 0.97 per run and ≥ 0.95 per question type. Answer correctness is reported for continuity only: its F1 term penalizes complete answers — among Round 3 graph-configuration answers with coverage ≥ 0.8, answer length and correctness correlate at $r = -0.45$ — and it averages only 0.76 on answers whose verse recall (≥ 0.9) and coverage (≥ 0.8) are both high.

C. A/B Infrastructure and Reproducibility

Four properties of the harness matter for the experiments that follow:

- **Per-request configuration**: `use_graph`, `semantic_only`, and `fusion_alpha` are request-level overrides, so the graph/no-graph/semantic A/B tests and α sweeps run against one container without restarts or rebuilds.
- **Fast retrieval loop**: with `retrieval_only=true` the backend skips generation; the 100-question retrieval suite completes in $\sim$15 minutes versus $\sim$2.5 hours for the full pipeline (36 min collection + 1 h 42 m RAGAS + 13 min coverage), with retrieval metrics verified *bit-exact* against the full pipeline in Rounds 0–2, while the intent classifier was inactive (with a sampling classifier the two can route a few questions differently).
- **Judge-visible context**: with `include_context=true` the backend returns, for every source, the exact context block the generator saw, and the judges score against that text — the repair of pitfall (v) above.
- **Re-scorable raw outputs**: every run archives raw responses (sources, strategies, scores per question); metrics can be recomputed from archives, and a comparison tool emits per-question deltas and hit-flip lists — the instrument behind every forensic table in Sections IX, X, and XI.

Retriever and encoder caveat. Two defects found by the October 2026 audit (Appendix C) apply to every round and remain in the reported numbers. First, retrieval was dense-only throughout, because the hybrid dense+sparse mode never executed (Section V-E). Second, the serving environment tokenized with transformers 5.0.0, which skips the NFKC normalization of the BGE-M3 and reranker tokenizers, whereas the index was built with version 5.3.0: full-width punctuation became the unknown token, and 499 of the 500 benchmark questions were encoded with input IDs that differ from the build-side encoding of the same text. The reranker was affected on both sides of its input — the query and the passage, whose RCUV text is punctuated in full width — and its query–passage pairs also used a non-standard separator (one end-of-sequence token where the model expects two). Correcting the serving-side tokenization (the embedder's query encoding and the reranker's query–passage input), measured in October 2026 on the 500 questions under that month's serving configuration (top-5 plus at most one appended passage), changed paired verse recall@6 by -0.003 (95% CI $[-0.013, +0.006]$) and MRR by $+0.005$ ($[-0.012, +0.023]$): a correctness fix without measurable retrieval effect. Since all arms within a round shared the same encoders, the Round 0–3 comparisons are reported as run.

VII. Round 0: Signal-Adaptive Retrieval vs. Semantic-Only

The first experiment round (May 2026) established the value of the architecture itself, before any knowledge-graph optimization.

A. Setup

Four configurations cross two retrieval modes with two answer models on identical code: {signal-adaptive (graph), semantic-only} $\times$ {Claude Haiku 4.5 [39] (commercial API), Gemma 4 E4B [38] (local, zero cost)}, with temperature 0.1 and top-5 context. The semantic-only baseline bypasses signal

TABLE VIII
ROUND 0 — SIGNAL-ADAPTIVE VS. SEMANTIC-ONLY (100 QUESTIONS, MAY 2026)

		Signal-adaptive		Semantic-only	
Category	Metric	Claude	Gemma	Claude	Gemma
Retrieval	Hit rate	0.960	0.950	0.810	0.810
	Recall@5	0.886	0.889	0.758	0.758
	MRR	0.805	0.802	0.647	0.647
Generation	Faithfulness	0.957	0.951	0.899	0.903
	Relevancy	0.815	0.893	0.709	0.774
	Context recall	0.797	0.791	0.680	0.677
	Coverage	0.750	0.722	0.618	0.628

Retrieval metrics under semantic-only are identical across answer models by construction (deterministic retrieval); small graph-mode differences arise because the answer LLM also performs intent classification, which shapes route selection and graph-traversal inputs. (With Gemma the classifier in fact returned nothing, so its routes are dictionary-only; Claude's classifier functioned and re-routed 8 of the 100 questions — Section V-C.) Faithfulness here and in Tables XII and XIV predates the judge repair of Section VI-B and understates it.

TABLE IX
ROUND 0 — ANSWER COVERAGE BY QUESTION TYPE (SIGNAL-ADAPTIVE)

Type	Claude	Gemma	Δ
VERSE_LOOKUP	1.000	0.975	−0.025
TOPIC	0.900	0.856	−0.044
PERSON	0.635	0.594	−0.041
EVENT	0.613	0.606	−0.007
GENERAL	0.601	0.580	−0.021

routing and all graph strategies, using pure dense retrieval. At this time point the entity-query strategy was *disabled*: an earlier deployment had found that cross-encoder reranking systematically displaced its candidates — surface-form scores beat entity anchors — so EQ was turned off pending a mechanism fix. This early incident is the first of the four incidents assembled in Section XIII.

B. Results

Three findings (Tables VIII and IX):

Signal-adaptive retrieval is the dominant quality lever. The architecture beats the semantic-only baseline on every metric, with decisive retrieval gains: hit rate +15 pp (0.81 → 0.96), recall@5 +13 pp, MRR +16 pp. The improvement propagates to generation (faithfulness 0.90 → 0.95, coverage 0.62 → 0.74, two-model averages) and is *model-independent*: Claude and Gemma improve by near-identical margins, isolating retrieval architecture as the causal factor. (The retrieval and coverage gains replicate on the same 100 questions in Round 3; the faithfulness gain is not confirmed under the repaired judge — Section XI.)

A local model matches the commercial API. With identical code — though not identical routing, since Gemma's classifier was silently inactive (Section V-C) — the two models differ by under 4 pp on six of seven metrics; only answer relevancy differs more (7.8 pp, in Gemma's favor — partly the noncommittal-refusal artifact of Section VI-B). Strengths are complementary (Claude leads faithfulness and coverage, Gemma relevancy); neither dominates. The practical consequence: the system runs at zero marginal cost with full data privacy, and the investment frontier is retrieval infrastructure, not model scale.

The hard question types pointed at the knowledge graph. VERSE (1.000/0.975 coverage) and TOPIC (0.900/0.856) are largely solved by exact SQL and semantic search; PERSON (0.635/0.594), EVENT (0.613/0.606), and GENERAL (0.601/0.580) lag. Person questions hinge on entity mentions and their (absent) coreference; multi-chapter narratives strain fixed-window retrieval; and at this time point the cross-reference network had only 916 edges to serve cross-book questions. Notably, PERSON was the *only* category where graph mode trailed the semantic baseline (hit 0.95 vs. 1.00) — a deficit that the audit traces to entity-layer damage rather than to routing. (Round 3 locates it in the graph strategy itself: with identical routing and no graph, all 20 original PERSON questions hit; Section XI-C.) These observations motivated the audit of Section VIII. That reading was taken on the legacy system in Round 0; on the rebuilt system, graph retrieval on targeted multi-hop questions showed no detectable gain against k-aligned non-graph controls; its one positive reading, on the development set on which the configuration was selected, did not hold on a second set (Section XII).

VIII. KNOWLEDGE-GRAPH QUALITY AUDIT: THE GRANULARITY-MISMATCH CAUSAL CHAIN

Guided by the hypothesis that "KG quality problems trace back to the original chunking design," we audited the entire construction pipeline on July 5, 2026, with every number re-verified live (Cypher queries, JSONL recomputation, line-level code reading). The hypothesis was confirmed — *but only after being made precise*: the damage was caused not by chunking parameters (512/768 tokens, 1-verse overlap are all defensible) but by the **silent inheritance of an embedding-oriented granularity structure by three downstream layers** (extraction, import, retrieval).

A. Eight Quantified Gaps

Table X lists the eight gaps with their live evidence. The three heaviest (#2, #3, and the split-pericope blindness described below) are all granularity-inheritance effects; the others are missing pipeline features (no coreference, no entity resolution, no alias generation) that chunk boundaries then amplify.

B. Six Mechanisms (M1–M6)

M1 — Extraction feed = embedding queue (structural root cause). The NER track consumed `embedding_queue.jsonl` directly: 2,610 pericopes + 431 chunks + 31,031 verse labels (10 of them ghost verses). This queue's granularity was designed for retrieval (split pericopes appear only as chunks; every verse appears individually); extraction inherited it unconditionally, so most NER mentions were emitted at verse granularity.

TABLE X
THE EIGHT AUDITED GAPS (JULY 5, 2026, PRE-REMEDIATION; ALL NUMBERS VERIFIED LIVE)

#	Gap	Live evidence	Mechanism
1	Empty aliases	35 of 9,122 entities (0.38%) have any alias; dictionary–graph naming mismatches make alias fallback dead code	pipeline gap
2	Silent mention loss	97,235 of 173,896 mentions (55.9%) are verse-level; the importer's MATCH finds no verse node, the MERGE silently no-ops, and the counter still reports "173,896 imported"	M1+M2
3	Retrieval-blind entities	1,474 entities (16%) have no pericope anchor (1,030 zero-anchor + 444 chunk-only); graph retrievers match only `(Pericope)-[:MENTIONS]`	M2, M3
4	Duplicate-name splits	878 names split across 1,900 nodes, all cross-type (e.g., the Ark appears as Group/Object/Person/Place); alias-based merging is inert because aliases are empty	pipeline gap × M5
5	Hollow event layer	96.2% of 1,712 events have no temporal/causal edge (45 edges by the audit's count); 67.9% lack a place and 65.6% lack any participant (construction-CI measurement)	pipeline gap × M3
6	Relation import rate 8.1%	85,439 candidate pairs → 6,958 imported; 77,953 unclassified retained with provenance	pipeline gap
7	Noise entities	the conjunction "dan" (but) indexed as the place Dan with hundreds of anchoring sources; "Jehovah" typed as Group; generic "day(s)" event with 413 mentions	pipeline gap
8	Upstream extraction quality	4B-parameter classifier; 33 single-character stopwords remove God/Spirit/Lord; descriptions are 100-char snippets (46% of NER-type entities have none), polluting entity vectors — an upstream cofactor of the May EQ displacement	pipeline gap

M2 — Import-layer granularity mismatch: 55.9% of mentions evaporate. Verse-level mentions have no corresponding Neo4j node (Section IV-D); the import MATCH silently no-ops and 97,235 mentions vanish while the log reports success. Recomputing offline showed the loss was fully recoverable: 100% of verse source IDs can be remapped to their parent pericope by stripping the verse suffix, which would add 5,853 unique (entity, pericope) anchor pairs and re-anchor 1,310 of the 1,474 blind entities (89%).

M3 — Two-layer misalignment: 18.3% of the text doubly blind. For the 169 split pericopes, NER-type entities (Person/Place/Group) anchor only at chunk/verse level, while *both* the graph retrievers and the relation-extraction pair miner match only pericope-level MENTIONS. Entities of these Old Testament narrative passages were therefore invisible to retrieval *and* excluded from relation extraction — precisely the person×event material of the hardest benchmark categories.

M4 — Chunk boundaries × absent coreference. The pipeline has no coreference resolution anywhere; mentions are located by literal string search, and the 1-verse overlap rarely contains the antecedent of "his father-in-law" or "he." Document-level relation extraction quantifies the ceiling this imposes: at least 40.7% of relational facts require cross-sentence evidence and 17.6% require coreference reasoning [6]. (We found no peer-reviewed quantification of overlap-size effects on extraction, and therefore rely on this indirect evidence.) The kinship edges missing for Jethro (Exodus 18) — where the text says "Moses' father-in-law" with the name far away — are the local instance.

M5 — Mixed granularity: duplicated and inconsistent extraction. The same text is extracted at up to three granularities; the same entity may be found at fine granularity and missed at coarse, with only toneless-pinyin surface forms as deduplication keys (which also *over*-merge homophones). Extraction granularity directly controls density — GraphRAG's ablation found nearly twice the entity references at 600-token versus 2,400-token chunks [3] — so mixing granularities guarantees inconsistency; and atomic-fact decomposition work confirms finer units extract more completely [34]. Cross-chunk relation loss is a known systematic failure of chunk-local extraction [33], and retrieval-augmented construction [31] exists precisely to re-assemble such context.

M6 — 512/768 tokens was never an extraction constraint. The 169 split pericopes have median length 1,353 characters (max 3,544) — far below the context degradation range documented by [27]. Extraction could simply have consumed whole pericopes; chunks should have remained a storage/retrieval concern. (The "storage unit ≠ extraction unit" separation is standard in production KG builders, e.g., Neo4j's LLM Graph Builder recombines chunks before extraction [40].)

C. Attribution and the Cross-System Survey

The audit's precise conclusion: *the chunking parameters are innocent; the unconditional inheritance of chunking granularity by extraction, import, and retrieval is guilty.* Independent of granularity, the missing-feature defects (no coreference, no entity resolution, no alias generation, snippet descriptions) have direct literature counterparts: type-aware coreference cuts node duplication by 33.28% [7]; removing coreference alone raises duplication by 28.25% while removing structured prompting mainly raises noise [8]; three-stage LLM coreference achieves -45.21% duplication with the benefit *growing* with document length [9] — longest for exactly the split pericopes our pipeline handles worst.

We also surveyed nine production GraphRAG codebases (Microsoft GraphRAG, LightRAG, nano-graphrag, fast-graphrag, Neo4j LLM Graph Builder, LazyGraphRAG, KAG, iText2KG, RAPTOR) at source level. Four regularities recur: (a) index-time deduplication is exact-name only, with approximate entity resolution deferred to post-hoc/incremental stages keyed on lexical-distance + embedding-similarity double signals; (b) merged surface forms are written back as aliases — which is *the* correct alias-generation mechanism our pipeline lacks; (c) multi-round gleaning re-extraction exists

TABLE XI
P0 DATA REPAIRS AND CONSTRUCTION-SIDE RESULTS (JULY 6, 2026)

Repair	Result
Verse-mention backfill	+5,853 pericope anchors; blind entities 1,474 → 164; entity anchor coverage 83.8% → 98.2% (prediction from the audit's offline recomputation matched exactly)
Import honesty	importer rewritten: verse remap + honest counters (silent-drop class eliminated for future rebuilds)
Alias injection	38 nodes gain dictionary aliases (the dictionary's coverage ceiling; mass aliases require entity resolution)
Noise gating	"dan" anchors 759 → 26 after classifying 1,882 mentions (44 geographic, 1,439 conjunction, 399 substring false hits inside Satan/Nathanael/Dedan/Abidan/Medan); 16 generic-noun events deleted across all three stores; Jehovah retyped Group→Person
Unclassified-relation rescue	Event-related pairs reclassified from the 77,953-pair pile: +5,641 PARTICIPATED_IN, +3,419 OCCURRED_IN (confidence 0.35, flagged as co-occurrence backfill); events with participants 34.4% → 84.0%, with places 31.8% → 75.5%
TSK import	CROSS_REFERENCES 916 → 250,418 (×273); retrieval-side expansion simultaneously switched to seed-support+votes ordering

only in the GraphRAG lineage [3] (the paper describes forced-continuation gleanings; cost-sensitive systems omit it); (d) mainstream systems extract at a *single* unified granularity — our three-granularity mix with no resolution stitching is the outlier and the fragmentation driver.

IX. ROUND 1: P0 REMEDIATION AND THE NEGATIVE RESULT

A. Six Data Repairs (No Re-Extraction)

P0 repaired data in place — days of work, no re-extraction — following the audit's cheapest-first ordering (Table XI). Construction-side quality indicators moved decisively; three of them were then tracked as construction-CI metrics of the legacy graph.

B. Evaluation Conditions

The 100-question graph-mode evaluation was re-run under conditions verified identical to the May baseline: same answer model (Gemma 4 E4B), same judge (Gemma 4 26B), same disabled entity query, and — verified question by question — the *same route distribution* (R1=20, R2=25, R3=21, R4=18, R5=10, R6=4, fallback=2; both runs under the silent classifier fallback of Section V-C). The only systematic difference was the state of the knowledge graph. Retrieval metrics are deterministic, so every delta below is exactly reproducible from archived raw responses.

C. Result: Construction Metrics Up, Retrieval Metrics Flat

Table XII is the paper's central negative result: after a completeness overhaul that raised entity anchor coverage by 14.4 points and multiplied cross-references by 273, **every deterministic retrieval metric was flat or slightly down**, and answer-side movements were inside judge noise. By category: VERSE and TOPIC unchanged at 1.00 hit rate; PERSON unchanged *question by question* (still 0.95, still losing to the semantic baseline's 1.00); EVENT dropped from 0.90 to 0.80 hit rate and its context recall from 0.608 to 0.546; GENERAL recall slipped from 0.654 to 0.638. The two indicators the audit had designated as success criteria — PERSON hit rate and EVENT context recall — both failed to move in the predicted direction.

TABLE XII
P0 NEGATIVE RESULT — OVERALL METRICS (100 QUESTIONS, GRAPH MODE)

Metric	Pre-P0 (May 16)	Post-P0	Δ
Hit rate	0.9500	0.9300	−0.0200
Recall@5	0.8892	0.8658	−0.0234
NDCG@5	0.8563	0.8496	−0.0067
MRR	0.8020	0.7978	−0.0042
Context recall (RAGAS)	0.7908	0.7767	−0.0141
Answer coverage	0.7223	0.7253	+0.0030
Answer correctness (RAGAS)	0.5874	0.5960	+0.0086
Faithfulness (RAGAS)	0.9513	0.9564	+0.0051

D. Mechanism Diagnosis (Question-by-Question)

Every regressed or unmoved question was traced through its archived sources. Three mechanisms, each verified rather than conjectured:

1) TSK votes are a double-edged sword: After TSK import, cross-reference expansion went from "most questions have no neighbors" to "any seed reaches ∼180 one-hop neighbors," and the expansion ranked candidates by community votes. But *high votes mean strong topical–theological association, not same-event narrative*. Table XIII shows the five forensic cases: in each, the displacing passages are topically related by the cross-reference logic yet wrong for the narrative question. The same mechanism *helped* topical aggregation questions (TOPIC coverage +0.031; the Sermon on the Mount correctly gained its Sermon on the Plain parallel), so the repair had to be category-sensitive re-weighting — not rollback.

2) The reranker monopolizes the final ranking: The final ordering at this time point was literally `sorted(passages, key=rerank_score)`: strategy priors and votes decided only *who entered the pre-rerank pool*, never the final rank. Consequently (a) construction improvements — correct anchors newly present in the pool — could not surface unless they also won the surface-form contest, and (b) construction side effects — topical noise whose surface forms match the query ("Paul," "resurrection," "prayer") — displaced correct answers directly. This is the same bottleneck that had forced EQ off in May (Section VII); the P0 evaluation is its second independent confirmation.

3) The benchmark reads only the head: The 1,310 entities re-anchored by the backfill are long-tail entities that had been extracted only at verse level; the 100 questions ask about head entities (Elijah, David, Moses) whose pericope anchors were already plentiful. **On this yardstick, construction-side quantity had already saturated** — the decisive premise for

TABLE XIII
FORENSICS — FIVE QUESTIONS WHERE VOTE-RANKED TSK EXPANSION DISPLACED CORRECT NARRATIVE ANCHORS (P0 EVALUATION)

Question	Pre-P0 sources (correct)	Post-P0 sources (displaced by)	Cross-reference logic of the intruders
EVENT_019 Paul's conversion	Acts 9 (Damascus road), Acts 22, Acts 26 (his own retellings)	Acts 13 ×2, Acts 28, Romans 1	Pauline mission-theme chains (pre-P0 this question's expansion found nothing; post-P0 it always fires)
EVENT_017 Resurrection day	Luke 24 (Emmaus), Mark 16	Mark 9, Matthew 17 (Transfiguration)	"Son of Man risen from the dead" prophecy chains
EVENT_015 Gethsemane	Luke 22	Romans 8, Ephesians 6	"prayer" theme chains
PERSON_011 Elijah & Elisha	1 Kings 19 (the call)	Matthew 27	"see whether Elijah will come" (crucifixion bystanders)
GENERAL_013 1 Cor. 15 vs. Genesis	1 Cor. 15, Isaiah 49	Acts 2, Romans 6, Revelation 21	resurrection/death theme chains

deferring further construction investment. This is the local, deployment-level face of the benchmark-dependence phenomena reported in [10], [11], [25].

4) Persistent failures (pre- and post-P0) typed by root cause: EVENT_008 (division of the kingdom) and EVENT_014 (the Last Supper) fail because the graph event nodes are named differently ("revolt of the northern tribes"; "Passover meal"), have no aliases, and EQ — the designed lexical bridge — was off. PERSON_004 (the roles of Moses, Aaron, and Miriam) fails structurally: the answer is scattered across chapters and co-occurrence density favors genealogies/rosters; no top-5 pericope set suffices. That question is typed to the aggregation layer (community summaries / RAPTOR trees [3], [22]), not to P0/P1. (Round 3 qualifies the hit-level part: without the graph, PERSON_004 retrieves Exodus 6–7 and hits, so its graph-mode miss is the displacement of Section XI-C; complete coverage still needs aggregation.)

E. Decision: Defer Re-Extraction, Fix the Ranking Layer

The audit's original roadmap continued with P1 (decoupled re-extraction, coreference, gleanings, entity resolution — week-scale). The diagnosis, however, showed that (i) P1's gains concentrate in the long tail, which this benchmark cannot read (mechanism 3); (ii) until the ranking layer transmits graph signals, further construction investment is absorbed by the reranker (mechanism 2); and (iii) the remaining blind-entity population had already shrunk to 164 (1.8%). P1 was therefore **deferred**, and three day-scale, evaluation-verifiable ranking-layer repairs were scheduled instead — with the explicit note that P1's value proposition was unchanged *as graph asset*; what failed was the claim that its effect would be observable on this benchmark. P1 was never carried out: the October 2026 rebuild stopped the knowledge layer at a name layer and a curated event registry, with no identity or relation layer, and Section XII reports what graph retrieval on that layer achieved.

X. ROUND 2: RANK-FUSION REMEDIATION

A. Three Repairs and Three Chained Fixes

Repair 1 — TSK de-noising by provenance-stratified weights. Expansion cap $30 \to 10$; TSK edges (votes < 999) declassed to hop weights 0.60/0.50 — deliberately below the semantic prior 0.7 — while manual edges keep 0.75/0.55; the R5 blanket weight uplift for expansion candidates was removed. Rationale: keep the data, demote its priority; category-sensitive by *provenance*, not by deletion.

Repair 2 — Head-event backfill and EQ re-enablement. Motivated by a coverage audit: 31 of the 54 dictionary event keywords matched *zero* Event node by name — the graph calls the Last Supper a "Passover meal." Eleven existing Event nodes received question-phrasing aliases and 18 curated Event nodes (56 MENTIONS edges, every anchor verified live) were added, synchronized across all three stores; the entity-query strategy was then re-enabled. This is a hand-curated, head-only instance of the P1 alias/entity-resolution program — days, not weeks.

Repair 3 — Linear rank fusion. The final ranking became $s_{\text{fused}} = (1-\alpha)\, s_{\text{rr}} + \alpha\, w$ (Section V-F), with α overridable per request for the ablation.

Full-suite re-evaluation then triggered three *chained* fixes, each from a per-question regression trace: (a) graph and EQ retrievers now remap chunk IDs to parent pericopes inside Cypher, fixing a duplicate-seat bug where `exo:29:0` and its chunk `exo:29:0:0` both occupied top-5 seats (and, as a side effect, restoring retrieval visibility to the 164 chunk-only entities); (b) the legacy EQ pin and graph-uncertainty pin were retired under fusion after they promoted $s_{\text{rr}} \approx 0.002$ noise to rank 1 on two GENERAL questions; (c) a keyword-exact event pin was added for the one failure class fusion cannot reach — Acts 9 narrates Paul's conversion calling him exclusively *Saul*, so every surface signal fails (reranker 0.074, rank 7) and only the curated dictionary→entity bridge can pin the anchor.

B. Results

Table XIV and Fig. 4 give the headline: hit rate 0.97, recall@5 0.906, context recall 0.830, coverage 0.766; EVENT hit rate 1.00 with context recall 0.728 — not merely recovering the P0 regression but surpassing the pre-P0 state on every EVENT indicator; and the margin over the semantic baseline widened to $+0.16$ hit rate / $+0.148$ recall@5. Four questions flipped from miss to hit — the two persistent pre-P0 failures (EVENT_008 via aliases+EQ, EVENT_014 via curated node + fusion) and the two TSK casualties (EVENT_017 via TSK declassing, EVENT_019 via the keyword-exact pin) — and all five forensic cases of Table XIII recovered, with EVENT_015

TABLE XIV
METRIC EVOLUTION ACROSS THE THREE TIME POINTS (100 QUESTIONS, GRAPH MODE, IDENTICAL DETERMINISTIC SCORING)

Metric	Pre-P0	Post-P0	Post-fusion	Δ[a]
Hit rate	0.9500	0.9300	**0.9700**	+0.0400
Recall@5	0.8892	0.8658	**0.9058**	+0.0400
NDCG@5	0.8563	0.8496	0.8653	+0.0157
MRR	0.8020	0.7978	0.8335	+0.0357
Context recall	0.7908	0.7767	**0.8297**	+0.0530
Answer coverage	0.7223	0.7253	**0.7664**	+0.0411
Answer correctness	0.5874	0.5960	0.6173	+0.0213
Faithfulness	0.9513	0.9564	0.9495	−0.0069
EVENT hit rate	0.90	0.80	**1.00**	+0.20
EVENT context recall	0.608	0.546	**0.728**	+0.182
PERSON hit rate	0.95	0.95	0.95	0
PERSON context recall	0.667	0.683	0.745	+0.062
TOPIC / VERSE hit rate	1.00	1.00	1.00	0
Margin vs. semantic (hit)	+0.14	+0.12	**+0.16**	—

[a] Post-fusion minus post-P0. Answer-side conditions identical to post-P0 (answer: Gemma 4 E4B; judge: Gemma 4 26B). Faithfulness flat across all three time points ⇒ answer-side gains derive from better context, not generation drift. Semantic-only baseline: hit 0.81, recall@5 0.7575. LLM-judged rows report the documented run of record; re-judging the *identical* retrieval outputs with the same judge shifts them within the noise band of Section VI-B (e.g., faithfulness 0.9495 vs. 0.9655) — one more reason all retrieval conclusions rest on the deterministic rows, which are bit-exactly reproducible. All faithfulness rows predate the judge repair of Section VI-B and are not comparable with Round 3.

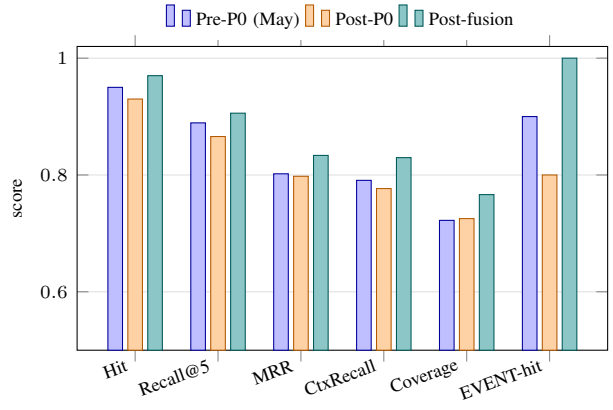


Fig. 4. The three-round shape of the evidence chain: the KG completeness overhaul alone (orange) reads flat-to-negative on a head-entity benchmark; unlocking the ranking layer (teal) recovers and surpasses every indicator.

regaining full recall (its Matthew and Luke anchors returned to the top-5). The quick retrieval loop and the full pipeline agree bit-exactly on every retrieval metric.

Two residuals are instructive rather than problematic. GENERAL NDCG sits 0.08 below post-P0 — traced per question to the *retirement* of pin-based artificial top-1 placement (the correct passages remain in top-5; the earlier perfect MRR was pin manufactured, so the "drop" is the removal of an artifact, not a regression). PERSON_004 remains the single PERSON miss, by design out of scope for ranking repairs (Section IX-D; Round 3 shows that its hit-level miss is caused by `graph_person` itself, Section XI-C).

TABLE XV
α ABLATION (RETRIEVAL-ONLY LOOP; ALL OTHER REPAIRS ACTIVE)

	$\alpha = 0$ (pure reranker)	$\alpha = 0.3$ (deployed)
Overall hit rate	0.97	0.97
EVENT hit rate	1.00	1.00
EVENT recall@5	baseline	+0.025
TOPIC MRR	baseline	+0.133
NDCG@5	baseline	−0.02

$\alpha = 0$ keeps the data repairs, chunk remap, pin changes, and keyword-exact pin; only the fusion term is disabled.

C. α Ablation: A Complementary Structure

Sweeping the fusion coefficient with everything else fixed (Table XV) decomposes the remediation:

- **Discrete repairs saturate hit rate.** At $\alpha = 0$, hit rate is already 0.97 and EVENT already 1.00: curated data (aliases, nodes) and the dictionary pin are responsible for *whether* the answer appears at all.
- **Continuous fusion recovers the tail.** $\alpha = 0.3$ adds EVENT recall@5 +0.025 — the multi-gospel parallel anchors of the Last Supper carry $s_{\text{rr}} \approx 0.01$–$0.02$ and only fusion returns them — and TOPIC MRR +0.133, at the cost of NDCG@5 −0.02 (high-prior, low-surface-form candidates advancing).
- **The trade resolves by downstream use.** All top-5 passages enter the generation context, so recall dominates internal ordering; $\alpha = 0.3$ was adopted.

The two mechanisms substitute for the reranker's surface-form monopoly at *different layers* — discrete repairs at the candidate/vocabulary layer, fusion at the scoring layer — and their effects are complementary rather than redundant. This decomposition is the constructive counterpart of the negative result: the same graph assets that read as zero in Section IX — together with the curated head events of Repair 2, which the $\alpha = 0$ column shows carry the hit-rate gain — produced the Table XIV gains once the ranking layer transmitted them. Like every number in this section, that reading was taken on the 100 questions the repairs were developed on. Section XI re-tests it on the 399 evaluable questions none of them saw and detects no graph gain there; on the rebuilt system, no gain from a graph operation was detected against k-aligned non-graph controls, and the one positive reading, on the development set on which the configuration was selected, did not hold on a second set (Section XII).

XI. ROUND 3: EXPANDED BENCHMARK AND GRAPH ABLATION

Rounds 0–2 — Round 0 (Section VII), the P0 round (Round 1, Section IX), and the rank-fusion round (Round 2, Section X) — were all read on the same 100 questions, and the Round 2 repairs were developed on exactly those questions: every regression trace of Section X was a trace of a benchmark question. Section IX had argued that this yardstick reads only the head; Round 2 added the complementary risk that its readings are in-sample. Round 3 (September 2026) therefore re-evaluates the deployed system end to end on the

500-question benchmark of Section VI-A, whose 400 new questions were written after the Round 2 repairs, with the repaired instruments of Section VI-B, and — for the first time — with the knowledge graph switched off while everything else is held fixed. Round 3 is the paper's last round on the legacy Neo4j system; graph retrieval on the rebuilt system is reported in Section XII.

A. Setup

Three configurations ran on one deployment and one code version (September 28–30, 2026), differing only in request-level switches (Section V-B):

- **Graph** — the full system of Section V (rank fusion, $\alpha = 0.3$).
- **No-graph** — identical signal routing, SQL strategies, book anchors, semantic (dense) search, reranking, fusion, and pins, but `use_graph=false` gates every Neo4j-backed strategy: the typed graph traversals, cross-reference retrieval and expansion, entity paths, and the entity query.
- **Semantic-only** — the Round 0 baseline: no routing, dense retrieval (top-20) and reranking only.

All three use Gemma 4 E4B as answer model (temperature 0.1), top-5 context, and Gemma 4 26B as judge, one run per configuration. Round 3 was not pre-registered: no primary metric, decision rule, or test family was fixed in advance, so all of its contrasts are exploratory. Four changes separate this deployment from Round 2. None of them changes the graph's content (a July 8 maintenance step only attached provenance tags to 106 existing manual edges and synchronized two manual event nodes into the other stores), although change (ii) changes which graph strategies run.

(i) `book_anchor` now retrieves per named book and suppresses same-book decoys (one anchor per book in the fused list; pins only for named books absent from the top-5); the earlier single OR-filtered search had let one book take all ten seeds. Unlike the Round 2 graph repairs, this fix was developed after a first evaluation of the 500 questions (July 12–14) and A/B-checked on them, so routing-side contrasts on the expansion set are not strictly out-of-sample.

(ii) The intent classifier was repaired. Gemma 4 E4B is a reasoning model whose hidden reasoning tokens exhausted the classifier's 256-token output budget; the empty reply failed JSON parsing, and the classifier silently fell back to `intent=topic` with empty entity and keyword lists. Replaying the signal detector with an empty classifier output reproduces the logged route of every question in every Gemma graph-mode run of Rounds 0–2 (100/100 in each 100-question run, 500/500 in a July 15 run on the expanded benchmark). Those rounds were therefore routed by the regex and dictionary signals alone, and strategies fed only by classifier output never ran: R5's entity traversal was invoked 0 times in the July 15 run and 73 times in Round 3. (The Claude configuration of Round 0 was unaffected; its classifier re-routed 8 of 100 questions.) With a 4,096-token budget, 114 of the 500 questions (22.8%) changed route relative to the July 15 run (48 fallback→R4, 27 R3→R5, 14 R6→R4, 12 fallback→R5, 7 R6→R5, 6 R4→R5; 7 of them among the original 100). Yet graph-configuration retrieval was essentially unchanged (verse recall@5 $0.751 \rightarrow 0.749$, anchor coverage@5 $0.785 \rightarrow 0.781$). The Ollama server that hosts the generator and the judges was upgraded alongside this fix.

(iii) Judges receive the generator's exact context blocks, faithfulness is scored by the repaired zh and strict judges, and verse-level retrieval metrics are computed for every run (Section VI-B). (iv) Merged verse labels ("29–30") no longer break exact-verse lookup.

Two operational facts qualify the comparison. The classifier samples at temperature 0.1, so 12 of the 500 questions (2.4%) were routed differently in the graph and no-graph runs although routing does not depend on `use_graph`. And one graph-run question (VERSE_LOOKUP_035) lost its only strategy to a network timeout; the archived aggregates score it as a miss (RAGAS skips it, $n = 499$), while all paired comparisons below exclude it.

B. Results

Tables XVI and XVII support three findings.

The head result replicates. On the original 100 questions the graph configuration reads hit rate 0.97 and a $+0.16$ margin over semantic-only (0.81) — the Round 2 headline, reproduced by an independent full run twelve weeks later and with a functioning classifier. The misses changed composition: GENERAL_008 now hits (re-routed to R5, whose newly active entity traversal reaches Matthew 21), EVENT_020 now misses (Section XI-C), and GENERAL_006 and PERSON_004 still miss.

Routing, not the graph, carries the margin. The no-graph configuration — routing, SQL, and book anchors without a single Neo4j call — reaches 0.94 on the original 100: $+0.130$ of the $+0.160$ (95% CI $[+0.070, +0.200]$), leaving the graph $+0.030$ ($[-0.020, +0.080]$). All 13 questions that no-graph recovers come from the 45 original questions routed R1 or R2, which cite a reference (exact-verse lookup $+5$, chapter fetch $+8$); on the other routes no-graph and semantic-only tie on hit rate. The graph's net share comes from EVENT questions (0.95 vs. 0.80): four head events whose dictionary keywords map onto graph event nodes (EVENT_008/011/014/019; three of them are among the four Round 2 flips) against one loss (EVENT_020); elsewhere one gain (GENERAL_008) offsets one loss (PERSON_004). Over all paired questions (499), no-graph beats semantic-only on every retrieval metric (hit rate $+0.050$ $[+0.030, +0.070]$, verse recall@5 $+0.040$, anchor coverage@5 $+0.043$) and on answer coverage ($+0.031$ $[+0.012, +0.051]$), whereas graph and no-graph tie on hit rate (0.944 each; seven flips each way, one of them the infrastructure failure). On the expansion questions the routing gain shrinks but survives (hit rate $+0.030$ $[+0.013, +0.050]$, anchor coverage@5 $+0.024$), largely because semantic-only itself does far better there (hit rate 0.915 vs. 0.810) — many expansion questions are single-anchor long-tail lookups that dense retrieval already finds — and because fewer of them cite a reference (33% are routed R1/R2, against 45% of the original 100).

TABLE XVI
ROUND 3 — THREE CONFIGURATIONS ON THE 500-QUESTION BENCHMARK (SEPTEMBER 2026)

	All 500			Original 100			Expansion 400		
Metric	Graph	No-graph	Semantic	Graph	No-graph	Semantic	Graph	No-graph	Semantic
Hit rate	0.944	0.944	0.894	0.970	0.940	0.810	0.938	0.945	0.915
Recall@5 (unit level)[a]	0.835	0.843	0.795	0.893	0.872	0.752	0.821	0.836	0.805
Verse recall@5	0.749	0.757	0.718	0.688	0.686	0.554	0.764	0.775	0.759
Anchor coverage@5	0.781	0.787	0.744	0.775	0.749	0.629	0.782	0.797	0.773
MRR	0.801	0.811	0.747	0.816	0.796	0.652	0.797	0.814	0.770
NDCG@5	0.836	0.845	0.779	0.856	0.842	0.692	0.831	0.845	0.800
Context recall	0.777	0.784	0.745	0.802	0.783	0.670	0.771	0.784	0.763
Answer coverage	0.669	0.684	0.653	0.729	0.735	0.636	0.655	0.671	0.657
Answer correctness[b]	0.618	0.632	0.604	0.589	0.590	0.511	0.625	0.642	0.628
Faithfulness (zh)	0.981	0.981	0.980	0.983	0.986	0.964	0.980	0.980	0.984
Faithfulness (strict)	0.980	0.979	0.979	0.989	0.980	0.965	0.977	0.978	0.983
Answer relevancy	0.865	0.878	0.840	0.868	0.892	0.778	0.864	0.875	0.856

Answer model Gemma 4 E4B, judge Gemma 4 26B, top-5, one run per configuration; "Original 100" are the Round 0–2 questions. Archived aggregates: the graph columns include the infrastructure failure VERSE_LOOKUP_035, scored as a miss (RAGAS $n = 499$); the paired comparisons of Table XVII exclude it. Faithfulness uses the repaired judges of Section VI-B and is not comparable with Rounds 0–2.

[a] Each gold pericope or chapter range counts as one unit; verse recall and anchor coverage are the verse- and chapter-level metrics.

[b] Reference only: its F1 term penalizes complete answers (Section VI-B).

TABLE XVII
ROUND 3 — PAIRED DIFFERENCES WITH 95% BOOTSTRAP CONFIDENCE INTERVALS

Contrast	Questions	Hit rate	Verse recall@5	Anchor cov.@5	Context recall	Answer coverage	Faithfulness (zh)
No-graph − semantic	All (499)	+0.050	+0.040	+0.043	+0.039	+0.031	+0.001
		[+.030, +.070]	[+.022, +.059]	[+.026, +.062]	[+.023, +.058]	[+.012, +.051]	[−.008, +.010]
	Original (100)	+0.130	+0.132	+0.120	+0.113	+0.099	+0.022
		[+.070, +.200]	[+.076, +.197]	[+.063, +.183]	[+.058, +.176]	[+.039, +.164]	[−.001, +.052]
	Expansion (399)	+0.030	+0.016	+0.024	+0.021	+0.014	−0.004
		[+.013, +.050]	[+.000, +.033]	[+.008, +.041]	[+.006, +.037]	[−.004, +.032]	[−.013, +.004]
Graph − no-graph	All (499)	+0.002	−0.007	−0.005	−0.006	−0.012	0.000
		[−.012, +.016]	[−.017, +.003]	[−.016, +.007]	[−.018, +.005]	[−.027, +.002]	[−.008, +.008]
	Original (100)	+0.030	+0.002	+0.026	+0.018	−0.006	−0.004
		[−.020, +.080]	[−.030, +.031]	[−.013, +.070]	[−.024, +.060]	[−.052, +.040]	[−.015, +.007]
	Expansion (399)	−0.005	−0.009	−0.012	−0.012	−0.014	+0.001
		[−.018, +.008]	[−.019, +.001]	[−.023, −.003]	[−.023, −.003]	[−.029, +.001]	[−.009, +.011]

Means of per-question differences; intervals (leading zeros omitted) from 10,000 bootstrap resamples with a fixed seed. VERSE_LOOKUP_035 (infrastructure failure) is excluded throughout. Intervals are per cell and uncorrected for multiplicity, and Round 3 was not pre-registered; for graph − no-graph, the upper end of an interval is the upper bound on the graph's average gain.

Multiplicity, graph − no-graph (question unit, $k = 5$): Holm's adjustment [41] over eight sign-flip tests (100,000 flips; a family defined by the October 2026 audit after the Round 3 results were known) — hit rate, verse recall@5, anchor coverage@5, and MRR, each on all 499 and on the 399 expansion questions — takes expansion anchor coverage@5 from $p = 0.013$ to 0.107, expansion MRR from 0.026 to 0.18, expansion verse recall from 0.092 to 0.55, and the other five cells to 1. A retrieval-only rerun of both configurations on October 3 gave expansion anchor coverage@5 -0.0103 $[-0.0215, +0.0001]$, $p = 0.060$. The MRR win:loss count (questions on which the graph configuration ranked the first gold passage higher vs. lower) is 12:35 on the expansion set (sign test $p = 0.001$; it survives a Holm adjustment applied to the sign tests of the same eight cells) and 13:32 in the rerun ($p = 0.007$); this statistic was chosen after the results had been seen.

On questions the graph repairs never saw, no graph gain was detected. On the 399 expansion questions, graph minus no-graph answer coverage is -0.014 $[-0.029, +0.001]$: the 95% CI upper ends are $+0.001$ for coverage and $+0.008$ for hit rate. Both come from one run per configuration and resample questions only; the coverage interval is about as wide as the ± 0.013 generation-and-judge noise measured below, so the retrieval bounds, which carry no judge noise (verse recall@5 upper end $+0.001$), are the firmer ones. Every retrieval point estimate is negative — hit rate -0.005, verse recall -0.009 $[-0.019, +0.001]$, anchor coverage@5 -0.012 $[-0.023, -0.003]$, MRR -0.015 $[-0.028, -0.002]$ — and so is the judged context recall (-0.012 $[-0.023, -0.003]$). These intervals are per cell and uncorrected. Holm's adjustment [41] over the eight sign-flip tests of hit rate, verse recall@5, anchor coverage@5, and MRR on all 499 and on the 399 expansion questions takes the anchor-coverage p from 0.013 to 0.107 and the MRR p from 0.026 to 0.18, so no retrieval mean difference survives it; a retrieval-only rerun of both configurations on October 3 reproduced the anchor-coverage direction only at $p = 0.060$ (Table XVII, notes). Only one harm statistic survives adjustment: the graph configuration ranked the first gold passage lower than no-graph on 35 expansion questions and higher on 12 (MRR win:loss 12:35, sign test $p = 0.001$, which survives a Holm adjustment applied to the sign tests of the same eight cells), and the rerun gave 13:32 ($p = 0.007$). That

statistic was chosen by an audit after the Round 3 results had been seen, so this harm signal is post hoc. Against semantic-only, the graph configuration's expansion advantage reduces to hit rate ($+0.025$ $[+0.005, +0.048]$) and, marginally, MRR ($+0.030$, interval lower bound ≈ 0), both uncorrected; the intervals for verse recall, anchor coverage, context recall, and coverage include zero. Descriptively, the losses concentrate in the person route: on the 55 expansion questions routed R3 in both runs, switching the graph on lowers verse recall from 0.692 to 0.627 and coverage from 0.606 to 0.567. By family (Table XIX) the largest losses are title_coref (-0.120 verse recall), multi_chapter (-0.049), and nt_quotes_ot (-0.031); typology is the one family with a sizable gain ($+0.117$, on $n = 12$). These route and family deltas are exploratory and descriptive: they carry no intervals and no multiplicity correction, and they rest on small subsets.

C. Mechanism: Wrong Graph Anchors Pass the Ranking Layer

Nothing in the ranking layer asks whether a graph anchor is right. The cross-encoder scores surface form, and the fusion prior scores the proposing strategy: on R3, R4, and R6 a typed-graph candidate carries $w = 0.85$–0.9 against 0.65–0.7 for a semantic one, so under $s_{\text{fused}} = (1 - \alpha)\, s_{\text{rr}} + \alpha\, w$ with $\alpha = 0.3$ (Section V-F) it outranks a semantic candidate unless the latter's reranker score is higher by roughly 0.06–0.11. When the anchor is right — the curated head events — injection plus prior is the intended rescue; when it is wrong, the same machinery lets it through. Table XVIII sorts the 253 questions that took the same route among R3–R6 in both runs by how many of the five context slots graph strategies hold in the graph run. Where they hold three, the graph run gains on the no-graph run ($+0.116$ hit rate: 5 net hits of 43, three of them on the original 100). Where they hold all five, it loses (-0.103 hit rate, 3 net of 29; -0.099 verse recall; -0.112 coverage), more on the expansion questions (-0.133, -0.130, -0.182) than on the original 100 (-0.071, -0.066, -0.038). The buckets are defined by the graph run's own output, so this is a descriptive decomposition, not a dose–response estimate.

Which part of the ranking layer lets the wrong anchors through? A passage receives the same reranker score in both runs, so its reranker score can be solved from its fused score and prior. On the 30 same-route questions where the graph run lost a hit or verse recall, the reranker alone already prefers every graph-held passage to every passage it displaced in 16 to 21 cases (depending on the exact prior of a few strategies): the damage is done by *injection* — graph strategies propose off-target pericopes that the surface-form reranker then favors. In 9 cases the prior is decisive for at least one displacement.

Three traces show both shapes. EVENT_020 ("What story does the parable of the prodigal son tell?") is sent to R4 by the repaired classifier, and `graph_event` injects two *other* parables — the wicked tenants (Luke 20, Matthew 21, Mark 12) and the sower (Luke 8) — which fill all five slots at fused scores 0.48–0.57. The correctly titled Luke 15 pericope scores 0.38 in the no-graph run, and the reranker alone already ranks the intruders higher (0.32–0.44 against 0.24; the prior adds only 0.045). Under the silent classifier of Round 2 the same question had fallen back to semantic search and hit. GENERAL_048 ("love your neighbor as yourself": Leviticus 19, Matthew 22, Romans 13) is the same injection case: R5's entity traversal, newly fed by the classifier, injects Pauline Acts pericopes that out-score Romans 13 on the reranker alone. PERSON_066 (Timothy's grandmother and mother, 2 Timothy 1) is the prior case: the weakest of the Acts pericopes that `graph_person` proposes for Timothy (Acts 20) and the gold 2 Timothy 1 pericope have nearly equal reranker scores (0.073 vs. 0.077), and the prior decides. The two shapes have different roots. In EVENT_020 the event lookup lands on the wrong entities, an alias and entity-resolution gap of the kind the audit measured (Table X). In the other two the entity is right, but its anchors — the pericopes that mention it — answer a different question: a gap in anchor precision that the audit did not list.

Head questions are not immune. PERSON_004, the graph's other head loss, is a prior case: `graph_person` fills all five slots with genealogies and rosters (1 Chronicles 6, 23, 24; Numbers 2, 26), displacing the two Exodus pericopes (the genealogy of Moses and Aaron, Exodus 6–7) that the no-graph run retrieves; and EVENT_020 is itself a head question. On the original 100, however, the graph's verse-recall losses (9 questions) are outweighed by its wins (17, all on PERSON and EVENT questions except GENERAL_008), whereas on the expansion set the losses outnumber the wins 22 to 13. Among the families, title_coref shows the largest loss (-0.120 verse recall, $n = 12$) and disambiguation none.

D. Faithfulness and the Measured Noise Floor

Faithfulness is flat at ≈ 0.98. Under the repaired judges, faithfulness is 0.980–0.981 (zh) and 0.979–0.980 (strict) in all three configurations; every configuration passes the gate (strict ≥ 0.97 per run, ≥ 0.95 per type), and no paired contrast differs from zero (Table XVII). The Round 0 reading that retrieval quality propagates into faithfulness ($0.90 \rightarrow 0.95$, Section VII) is therefore not confirmed under the repaired judge: across all paired questions the no-graph minus semantic-only contrast is $+0.001$ $[-0.008, +0.010]$; on the original 100 a $+0.022$ tendency remains, and its interval $[-0.001, +0.052]$ just includes zero — and cannot exclude a gap of the Round 0 size. With citation headers visible, the generator is about equally faithful to whatever context it receives, and retrieval quality surfaces in coverage instead (original 100: 0.636 semantic-only, 0.735 no-graph, 0.729 graph).

The noise floor, measured. Retrieval on R1, R2, and the fallback never touches Neo4j, so 296 questions received byte-identical contexts in the graph and no-graph runs (all 177 R1/R2 questions, 59 on the fallback, and 60 graph-route questions whose top-5 the graph left unchanged). Their answer-side differences are pure generation-sampling and judge noise: answer coverage moves by a mean $|\Delta|$ of 0.060 (14% of the questions by ≥ 0.2; mean $\Delta = -0.002$), answer correctness by 0.074, faithfulness by 0.028, and context recall — judged on the context alone — by 0.003. With a standard deviation of 0.131 for the per-question coverage difference, a paired

TABLE XVIII
ROUND 3 — GRAPH MINUS NO-GRAPH BY NUMBER OF GRAPH-HELD CONTEXT SLOTS

	All (253)					Original (55)				Expansion (198)			
Graph slots in top-5	n	ΔHit	ΔVR	ΔAC	ΔCov	n	ΔHit	ΔVR	ΔCov	n	ΔHit	ΔVR	ΔCov
0	43	0.000	0.000	0.000	+0.034	5	0.000	0.000	0.000	38	0.000	0.000	+0.038
1	53	−0.019	+0.001	−0.011	+0.002	7	0.000	−0.002	+0.021	46	−0.022	+0.002	−0.001
2	57	0.000	−0.015	+0.002	−0.030	7	+0.143	+0.057	+0.012	50	−0.020	−0.025	−0.036
3	43	+0.116	+0.020	+0.018	+0.013	12	+0.250	+0.059	−0.017	31	+0.065	+0.005	+0.025
4	28	0.000	−0.017	−0.026	−0.047	10	0.000	+0.003	−0.015	18	0.000	−0.028	−0.065
5	29	−0.103	−0.099	−0.061	−0.112	14	−0.071	−0.066	−0.038	15	−0.133	−0.130	−0.182

Questions that took the same route among R3–R6 in both runs (infrastructure failure excluded). A slot is graph-held if its passage was proposed by a typed graph traversal, cross-reference retrieval or expansion, an entity path, or the entity query. VR = verse recall@5, AC = anchor coverage@5, Cov = answer coverage. With no graph-held slot the retrieval metrics are identical in both runs (42 of the 43 contexts are byte-identical), so that row's coverage delta is essentially generation and judge noise.

TABLE XIX
ROUND 3 — RETRIEVAL BREADTH AND ANSWER COVERAGE BY QUESTION TYPE AND FAMILY

		Verse recall@5			Anchor coverage@5			Answer coverage			
Type / family	n	G	NG	S	G	NG	S	G	NG	S	Δ_{VR}
VERSE_LOOKUP (all)	100	0.950	0.960	0.850	0.950	0.960	0.850	0.854	0.876	0.817	−0.010†
original	20	1.000	1.000	0.750	1.000	1.000	0.750	0.940	0.904	0.750	0.000
longtail_book	48	0.979	1.000	0.896	0.979	1.000	0.896	0.878	0.925	0.878	−0.021†
head_cold	14	1.000	1.000	0.929	1.000	1.000	0.929	0.893	0.905	0.833	0.000
paraphrase	18	0.778	0.778	0.778	0.778	0.778	0.778	0.662	0.690	0.718	0.000
TOPIC (all)	100	0.815	0.827	0.742	0.862	0.870	0.808	0.704	0.704	0.638	−0.012
original	20	0.964	0.964	0.670	1.000	1.000	0.750	0.906	0.900	0.688	0.000
longtail_chapter	35	0.947	0.947	0.900	1.000	1.000	0.971	0.704	0.734	0.684	0.000
multi_chapter	25	0.373	0.422	0.395	0.449	0.479	0.470	0.454	0.447	0.418	−0.049
head_cold	20	0.988	0.988	0.971	1.000	1.000	1.000	0.815	0.775	0.785	0.000
PERSON (all)	100	0.590	0.614	0.615	0.646	0.681	0.678	0.570	0.599	0.595	−0.025
original	20	0.418	0.410	0.405	0.627	0.631	0.631	0.629	0.623	0.600	+0.008
longtail_person	30	0.852	0.879	0.891	0.886	0.908	0.908	0.733	0.740	0.755	−0.027
disambiguation	12	0.440	0.440	0.440	0.385	0.385	0.385	0.345	0.403	0.391	0.000
title_coref	12	0.505	0.624	0.624	0.514	0.625	0.625	0.502	0.573	0.583	−0.120
trajectory	18	0.337	0.348	0.361	0.417	0.469	0.480	0.422	0.467	0.472	−0.010
split_pericope	8	0.954	0.979	0.924	0.896	0.958	0.896	0.588	0.638	0.588	−0.024
EVENT (all)	100	0.734	0.734	0.738	0.840	0.813	0.813	0.677	0.671	0.633	0.000
original	20	0.431	0.404	0.398	0.742	0.604	0.604	0.670	0.640	0.585	+0.028
longtail_event	34	0.948	0.946	0.960	0.985	0.985	0.985	0.724	0.741	0.694	+0.002
temporal	16	0.418	0.442	0.446	0.531	0.536	0.536	0.469	0.469	0.425	−0.024
surface_form	16	0.741	0.755	0.755	0.823	0.823	0.823	0.625	0.619	0.606	−0.014
parable	14	1.000	1.000	1.000	1.000	1.000	1.000	0.871	0.836	0.821	0.000
GENERAL (all)	100	0.655	0.651	0.644	0.605	0.612	0.573	0.542	0.570	0.580	+0.004
original	20	0.627	0.652	0.545	0.508	0.510	0.408	0.499	0.609	0.556	−0.025
prophecy_fulfillment	26	0.683	0.697	0.677	0.647	0.679	0.603	0.498	0.517	0.552	−0.014
nt_quotes_ot	16	0.622	0.653	0.713	0.492	0.544	0.544	0.486	0.527	0.566	−0.031
typology	12	0.620	0.503	0.636	0.639	0.583	0.674	0.667	0.642	0.667	+0.117
parallel_account	14	0.815	0.802	0.802	0.815	0.810	0.810	0.636	0.650	0.671	+0.013
theme_thread	12	0.529	0.516	0.468	0.549	0.528	0.444	0.550	0.508	0.508	+0.013

G = graph, NG = no-graph, S = semantic-only; Δ_{VR} = G − NG verse recall@5, descriptive (no intervals, no multiplicity correction). "Original" rows are each type's 20 Round 0–2 questions; head_cold spans two types and is split by type. Archived aggregates: †the difference is entirely the infrastructure failure VERSE_LOOKUP_035 (R1, R2, and the fallback never call Neo4j, and all other questions on those routes retrieved identically in both runs).

difference of two 400-question coverage means carries about ± 0.013 of pure noise at 95% confidence — the size of the graph − no-graph coverage difference above (the retrieval metrics carry no judge noise), which is why every contrast is reported paired and with an interval. The zero-slot row of Table XVIII shows the same floor directly: identical retrieval metrics, $+0.034$ coverage.

E. What Round 3 Changes

Four revisions to the reading of Rounds 0–2 follow. First, the $+0.16$ head margin is real, but about four-fifths of it belongs to signal routing — all 13 original questions that no-graph recovers from semantic-only come from exact-verse (R1) and chapter (R2) lookup — and the remaining $+0.03$ is not distinguishable from zero (95% CI upper bound $+0.08$). Second, the graph's point estimate on the original 100 comes from head events whose graph anchors were built or verified

on the evaluation questions themselves; it does not transfer to the 399 evaluable questions written afterwards, where no gain is detected (answer coverage upper bound $+0.001$) and the graph more often lowers than raises the rank of the first gold passage. Third, the bottleneck has moved: Rounds 1–2 found a ranking layer that could not *transmit* graph signal; Round 3 finds one that cannot *discriminate* it — neither the reranker nor a fixed strategy prior carries per-query evidence of anchor precision. Fourth, the long-tail families — the questions that Section IX argued the head benchmark could not read — show no graph benefit either: identical hit rates for longtail_person, longtail_event, and longtail_chapter, and verse recall -0.027, $+0.002$, and 0.000, because dense retrieval already resolves single-anchor long-tail questions. Head saturation explains why P0 read zero on the original 100, but in this architecture a long-tail benchmark does not reveal hidden P0 value either.

At the time, the natural next repairs were anchor-confidence gating (admitting or weighting graph candidates by entity-resolution confidence, with penalties for hub entities and generic events), per-strategy slot quotas, query decomposition for multi-anchor questions, and a re-sweep on the 500 of the $\alpha = 0.3$ of Section X-C, which had been chosen on the original 100. The legacy graph was retired in October 2026, and Section XII reports graph retrieval on the rebuilt system (question sets in Section XII-A). The rebuilt knowledge layer has no identity or relation layer, so there is no entity-resolution confidence to gate on, and its ranked pool holds no graph candidates (a curated event lookup lane appends at most one passage after the top-k), so slot quotas and an $\alpha = 0$ run can no longer apportion graph damage. Where graph signals could still act, no gain was detected against k-aligned non-graph controls; the one positive reading, on the development set on which the configuration was selected, did not hold on a second set. Fusing PPR or name-layer ranks into passage selection lowered the number of missing hops recovered in a per-hop probe on the multi-hop set OLD31 (19 and 14 of 58, against 21 for the reranker alone; descriptive). Query decomposition, which is not a graph operation, gained $+0.101$ in hop coverage over a deeper dense pool on the seen multi-hop development set NEW82 (exploratory) but missed its pre-registered bar on the 204 multi-anchor questions of the benchmark as evaluated on the rebuilt system (GT v2; $+0.0174$ $[-0.0025, +0.0380]$ in anchor coverage@6). Giving the decomposer PPR-ranked names instead of non-graph names changed hop coverage on NEW82 by -0.017 (upper bound $+0.023$). Both lines were stopped by the pre-registered rule; this is not evidence of absence (Section XII-D).

XII. Graph Retrieval on the Rebuilt System: No Gain Detected

The October 2026 rebuild, which followed a data audit, retired the legacy graph of Sections IV–XI rather than rebuilding it (Appendix C). We then tested whether graph operations that can be built on the rebuilt knowledge layer improve multi-hop retrieval, pairing each graph arm with a non-graph arm that reads the same number of passages (k-aligned). None showed a detectable gain outside the development set, at the production k, or against a non-graph twin: the one positive reading, dense-seeded PPR at $k = 10$ on the set on which it was selected, is post hoc and did not hold on a second set. This section therefore reports upper bounds and three mechanisms consistent with the null. No comparison is confirmatory: the question sets are small, written by an LLM, and partly used during development.

A. Setup

Rebuilt knowledge layer. There are two rebuilt releases, Rebuild-1 (`b20261008_6daa4f31`) and Rebuild-2 (`b20261008_e05d3e55`); all rebuilt-system experiments below use the data layer of Rebuild-2. Neither has Neo4j, an entity-identity layer, or a relation layer. What remains is the L0 name layer (2,926 names) and the L1 curated event registry (31 events, 33 in Rebuild-1, with 179 hand-curated anchors); the query-alias table has 17 entries. Online, the name layer serves only as routing vocabulary for person and place signals, and the registry enters retrieval only as a *curated event lookup lane*: a literal trigger match that appends at most one anchor passage after the finished top-k, takes no part in ranking, and picks the first anchor the core lacks, regardless of what the question asks. It is a lookup, not a graph operation.

Graph and retrieval stage. The PPR arms use an unweighted bipartite graph linking the 2,773 passages to 2,929 body-text name keys by 16,616 edges; 635 passages carry no name. The keys are the normalized proper-name spans of the passage bodies (hence 2,929, not the 2,926 names of L0), so different people with the same name share a node. GD and the October 18 arms run offline at the retrieval stage: exact float64 cosine over the 34,058 BGE-M3 records (a passage takes its best record's score), then the production cross-encoder (BGE-reranker-v2-m3, whole passage, maximum length 512), which scores each candidate against the question or, in the sub-query arms, against the sub-query. Routing, book anchoring, and the event lane are not involved, so absolute levels are not production levels.

Arms (names as in Table XX).

- **GD** (our label for single-shot, dense-seeded PPR; October 14–15). The pool is the dense top-20 plus the 10 highest-PPR passages outside it, with PPR seeded by the dense top-10 (weights $1/\mathrm{rank}$, restart probability 0.5, 40 iterations); the control C fills the same 30-passage pool with dense ranks 21–30. The cross-encoder picks the top-k ($k = 10$ primary, $k = 5$ the production k). PPR is the operator family of HippoRAG 2 [20]; GD does not reproduce that system.
- **Appended-passage arms** ("realkg"; OLD31, answer level). Every arm keeps the production top-5 (D5) and appends $m \geq 1$ passages per question (49 in total; KIN and L0 append fewer where they have too few candidates); the k-aligned dense control DK is production retrieval with top-$k = 5 + m$. KIN walks 1–2 hops over name-level kinship triples that a prototype extracted from the rebuilt text (not a production layer); L0 takes the passages whose body names include a name linked in the question; both rerank against the question. ORACLE

appends gold passages for the missed hops and FACTS gives the generator an author-written fact chain; both are ceilings, not methods.

- **Iterative arms** ("iterq"; OLD31, answer level). At each of m steps Gemma 4 E4B writes a sub-query s_t from the question and the current context, and one passage is appended: IT-D draws it from the dense candidates of s_t, IT-L0 from the L0 passages of the names linked in s_t.
- **Sub-query arms of October 18** (NEW82, retrieval). Every arm starts from C5, the five passages C ranks first, and appends one passage outside C5 ($k = 6$; C6 and C7 are C's top six and seven). DEEP adds the cross-encoder's best passage for the question among its dense top-50 outside C5, with no LLM. QD (query decomposition) has Gemma 4 E4B (temperature 0) write one sub-query from the question and C5 and adds the cross-encoder's best dense candidate for it. QD+G adds to the same prompt a line of five names ranked by PPR on the L0 graph seeded with the C5 passages; its non-graph twin QD+N uses the same template and name formula on the question's dense ranking; QD+ORACLE inserts eligible gold bridge names (a diagnostic). QD-type arms average three decompositions. Here IT-D (unlike the OLD31 arm of the same name) runs two context-conditioned sub-query steps after C5 ($k = 7$); IT-GD adds PPR candidates at each step, seeded with the dense top-10 of the current sub-query s_t, not with C5.
- **GN-41** (legacy Neo4j KG, context). KG bridge names added to the query, and KG facts given to the generator, each against the same pipeline's LLM query rewrite (top-5).

Attribution rule. Written into the pre-registration of the October 18 run before it started; for the earlier experiments (GD, realkg, iterq, and the event lane) we apply it retrospectively, after their results were known. Under it: only PPR or a traversal of two or more hops counts as a graph operation; a one-hop lookup is descriptive; LLM query rewriting is not a graph operation; and a graph arm must beat both the k-aligned dense arm and its non-graph twin. GD, KIN, QD+G, and IT-GD are graph operations; L0, IT-L0, and the event lane are lookups. Its main retrospective effect is to class the event lane's in-sample, k-aligned retrieval gain on the 23 questions where it fired (Section XII-B) as a lookup result, not as graph evidence.

Question sets. *NEW82*: 82 multi-hop questions with 223 hops (36, 34, 11, and 1 questions of two to five hops) in eight categories, kept from 90 LLM drafts. Against a host-side approximation of the production top-5, 102 hops in 63 questions are missing. Questions that share a gold passage form union-find clusters: 40, the largest with 12 questions. NEW82 is the development set (GD was chosen on it), so every result on it is exploratory. *OLD31*: the 31 of 38 earlier LLM-written questions on which frozen production top-5 retrieval missed at least one hop; 101 hops, 58 missing, 20 clusters (largest 9). Selected for production failure, its baselines are low by construction, and it is only partially blind: 16 of its questions share gold passages with NEW82, GD's seed form was chosen after OLD31 pool results, and realkg, iterq, and the per-hop probes had used it. *GN-41*: 41 questions on the legacy KG (13 title-bridge, 11 disambiguation, 8 kinship two-hop, 9 aggregation). *GT v2*: the 500 benchmark questions of Section VI-A as evaluated on Rebuild-2; 204 have more than one reference (*multi-anchor*). This subset had already been analyzed on the rebuilt system before the multi-anchor sub-query step was designed, and that step used a host-side dense reimplementation, not production retrieval.

Why not the 500-question benchmark. It is powered to detect broad harm from a graph, not the small gains a graph could add on it. 275 questions (55%) state the answer location (167 explicit citations, 108 single-anchor lexical lookups); after an LLM-assisted relabeling one question has multi-hop as its primary type and 15 need two or more hops; and if the legacy KG had served perfectly every non-curated question it could, the ideal verse-recall gain ($+0.018$, at most $+0.030$) would be only 1.2–2 times its minimum detectable effect (0.0146), and any realistic repair would deliver a fraction of that ideal.

Metrics and statistics. Retrieval is scored by hop coverage $f(q) = \frac{1}{H}\sum_{h=1}^{H} \mathbf{1}[\text{gold}_h \cap \text{top-}k \neq \emptyset]$ for a question with H hops. OLD31 answers are graded blind by two Claude raters and averaged over raters and three generations (linear κ 0.98, which measures agreement, not validity); GN-41 uses a deterministic answer-key score and GT v2 the metrics of Section VI-B. On NEW82, and for GD on OLD31, the unit is the cluster: $\bar{\Delta}$ is the mean of the clusters' mean paired differences (zeros included), W:L counts clusters with positive and negative means, the test is an exact two-sided cluster sign-flip, and the 95% CI is a percentile bootstrap over clusters (10,000 resamples, seed 20261014). realkg, iterq, GN-41, GT v2, and Round 3 use questions as the unit (paired bootstrap, 10,000 resamples), with sign and exact Wilcoxon tests (realkg, iterq), a bootstrap p (GN-41), or sign-flip and sign tests (the three-arm GT v2 run, Round 3), and Holm's adjustment [41] where a family was declared. The *upper bound* is the upper end of the 95% CI.

Evidence classes. Nothing here is confirmatory. *Pre-registered stopping rule*: GD's first pre-registered direction check on OLD31 (V1), rules G, O, and I of October 18, and the multi-anchor QD bar of its step 2, each committed to the project repository before its data were read (commits in Appendix B; no external registry); a stop is "stopped by the pre-registered rule; this is not evidence of absence", and since rules G, O, and I ran on the development set, a pass would have been exploratory only. *Exploratory (development set)*: everything on NEW82, realkg and iterq on OLD31, and GN-41, whose stop rule was written after generation. *Descriptive*: the event lookup lane, the release comparison, and the per-hop probes. Legacy Round 3 was not pre-registered.

Who wrote the questions. NEW82, OLD31, and GN-41 were written by an LLM (Claude) and never reviewed by a human. An independent LLM audit of NEW82 rated 33 questions ok, 33 minor, 11 major, and 5 critical before the flagged items were revised; three OLD31 questions score 0 in all six realkg arms, FACTS included, which suggests defective items; and the OLD31 answers were graded by two Claude

raters. This limits external validity (Section XIV).

B. Results

Single-shot PPR (GD). On NEW82 at $k = 10$, GD beat C in all 11 discordant clusters ($+0.063$ $[+0.028, +0.104]$, exact $p = 1/1024$), but GD was selected after the fact from about ten settings on this set, so we do not read that interval as a bound. At the production $k = 5$ no gain was detected (8:2 clusters, upper bound $+0.054$), and against a pool of the dense top-20 alone GD reads $+0.005$ $[-0.026, +0.034]$. The pre-registered direction check V1 on OLD31 found 2:2 with 4 of 20 clusters discordant: stopped by the pre-registered rule; this is not evidence of absence. At the discordance rate assumed in the pre-registration (0.222) the rule stops with probability 0.552 under the null, 0.190 at a cluster win rate of 0.7, and 0.072 at 0.8, so the stop argues against a cluster win rate as high as 0.8 but says little about moderate or small effects. The blind held-out set (120 questions) was never generated, the confirmatory comparison never ran, and the answer layer was never measured.

Kinship walk and name lookups (OLD31). KIN and L0 tie with DK on most questions (W:T:L 3:26:2 and 4:24:3); the effects detectable with 80% power are 0.085 and 0.121. In the iterative pipeline the name-layer pool reads -0.105 against the dense pool (3:18:10, sign $p = 0.092$); its pre-coded rule did not judge IT-L0 worse, so we read it as no gain detected, direction negative. The answer-level headroom is large, although by mechanism (C) little of it is reachable only through the graph: from DK's mean of 0.134, gold passages raise correctness to 0.481 and a full fact chain to 0.637, whereas KIN and L0 reach 0.129 and 0.151.

PPR names for an LLM decomposer (October 18). QD+G beat DEEP (16:2 clusters, $+0.082$ $[+0.038, +0.131]$), but so did its non-graph twin (QD+N, 18:2, $+0.099$ $[+0.050, +0.150]$). Rule G required QD+G to pass a gate (cluster W $\geq$ 2L, W$-$L ≥ 4, and $\bar{\Delta} > 0$) against both QD+N and DEEP, and to beat plain QD on $\bar{\Delta}$ and W:L; it lost to QD+N 6:8 and to QD 6:10. Rule G failed: stopped by the pre-registered rule; this is not evidence of absence. The 6:8 is not a near miss: with 12–16 discordant clusters the gate passes with probability 0.92–0.93 at a cluster win rate of 0.8, against 0.11–0.19 under the null. Gold bridge names did not help QD either (rule O: 5:4), although only 24 of 82 questions had eligible names, and adding PPR candidates to iterative retrieval gave $+0.010$ with 2:0 clusters (rule I; 38 of 40 clusters tied, so the interval $[0.000, +0.027]$ rests on two clusters), with identical final contexts on 72 of 82 questions. Rule I did not itself stop IT-GD; the authors' pre-run ruling specified that a failure of rule G closes every retrieval variant on the current graph layer, IT-GD included (Section XII-D). Mean $f(q)$@6 was 0.685 for QD, 0.665 for QD+G, 0.684 for QD+N, and 0.581 for DEEP. No arm used a decomposer fallback (0 of 246 decompositions each), and the cross-encoder scores of 32 fixed query–passage reference pairs reproduced exactly before the run and after each step.

Lookups and releases (GT v2). In the three-arm GT v2 run (full pipeline, the same without the event lane, and semantic-only; "eval3c" in the artifacts), switching the event lookup lane on moved answer coverage by $+0.0089$ (Holm $p = 0.137$ over that run's three primary comparisons). The lane fired on 23 questions, and contexts were identical on 473 of 500 questions; with one generation per arm, most changed per-question outcomes therefore reflect answer noise, not the lane. The comparison is not k-aligned (fired questions read six passages, not five). On the 23 the change is $+0.053$ $[-0.040, +0.163]$, against an observed minimum detectable effect of 0.155; a k-aligned retrieval check on them (verse recall $+0.163$ $[+0.069, +0.266]$) is in-sample and measures a lookup, not a graph. On NEW82 the lane fired once. Between releases, top-k changed on 24 of 500 questions (4 in an A/A rerun), and answer coverage on 200 questions moved by $+0.0019$ $[-0.0138, +0.0172]$, less than the A/A drift of $+0.0149$.

Legacy graph. Under multiplicity correction, Round 3 detected no gain (Section XI): the expansion anchor-coverage loss (-0.012, raw $p = 0.013$) has Holm-adjusted $p = 0.107$ over the eight-test family, and $p = 0.060$ in a retrieval-only rerun. The harm direction is consistent, but only the first-gold-rank win:loss counts survive a Holm adjustment applied to the sign tests of the same eight cells (MRR 12:35, sign $p = 0.001$), with the same direction in the rerun (13:32, $p = 0.007$; the rerun is not part of that family), and that statistic was chosen after looking. On GN-41 neither KG arm beat the LLM rewrite (Holm $p = 1$ for both) on a set near saturation (no-graph key score 0.866); its stop rule was written after generation, so the result is a null, not a pre-registered stop.

Non-graph context. The largest retrieval gain on NEW82 is LLM sub-query rewriting without a graph (QD $-$ DEEP, 19:2 clusters). It comes from the rewritten query, not from the extra slot: QD $-$ C6 is $+0.099$, DEEP $-$ C6 -0.002 $[-0.017, +0.015]$. It did not carry over to the 204 multi-anchor GT v2 questions (29:20), which missed the bar of $\geq +0.03$ with a CI lower bound above zero: stopped by the pre-registered rule; this is not evidence of absence. A blind confirmation of QD was not run, by decision. On OLD31, IT-D $-$ DK has Holm $p = 0.123$, and about 80% of it comes from the 18 questions that needed one appended passage: a single context-conditioned rewrite, neither iteration nor a graph.

C. Failure Mechanisms

Three mechanisms, (A)–(C), are consistent with the absence of a detectable gain. They are descriptive readings of the same development data, not tested causal hypotheses.

(A) Selection. The cross-encoder scores a passage only against the question or sub-query. Where it fills ten slots against the full question, as in GD, it keeps missing hops from the graph arm's pool at about the rate it keeps them from the control's (31 of 59 vs. 17 of 35; Table XXI). Where it fills only m appended slots or scores against a per-hop sub-query, graph-proposed bridge passages reach the pool but rarely survive: the L0 pool held 37 of 58 missing hops and 8 were kept, and in the per-hop probe, gold passages brought in by the graph were kept 0 times. In that probe, bridge passages scored mostly 10^{-3}–10^{-2}, near the bottom of their pools (e.g., `mal.1.1` ranked 20 of 20 at 0.0009, `1ki.16.15` 15 of 21 at 0.011). Of

TABLE XX
GRAPH, LOOKUP, AND NON-GRAPH COMPARISONS ON THE REBUILT AND LEGACY SYSTEMS — PAIRED DIFFERENCES AND UPPER BOUNDS

Comparison	Set (n; unit)	Metric @k	$\bar{\Delta}$	95% CI	Upper bound	Class
(a) Graph operations on the rebuilt layer (PPR or ≥2-hop traversal)						
GD (dense-seeded PPR) − dense-30 (C)	NEW82 (82; clusters)	$f(q)$@5	+0.020	[−.015, +.054]	+0.054	exploratory (development set)
GD − C, primary k (GD selected on this set)	NEW82 (82; clusters)	$f(q)$@10	+0.063	[+.028, +.104]	—[c]	exploratory (development set)
GD − C, V1 direction check	OLD31 (31; clusters)	$f(q)$@10	sealed[a]	—	—	pre-registered stopping rule
Kinship walk (KIN) − k-aligned dense (DK)	OLD31 (31; questions)	answer correctness, $5{+}m$	−0.005	[−.065, +.048]	+0.048	exploratory (development set)
PPR name hints (QD+G) − non-graph twin (QD+N), rule G1	NEW82 (82; clusters)	$f(q)$@6	−0.017	[−.060, +.023]	+0.023	pre-registered stopping rule
QD+G − plain sub-query (QD), rule G3	NEW82 (82; clusters)	$f(q)$@6	−0.019	[−.050, +.013]	+0.013	pre-registered stopping rule
Iterative + PPR candidates (IT-GD) − IT-D, rule I	NEW82 (82; clusters)	$f(q)$@7	+0.010	[.000, +.027]	+0.027	pre-registered stopping rule
(b) Lookups and releases (not graph operations under the attribution rule)						
Name lookup (L0) − DK	OLD31 (31; questions)	answer correctness, $5{+}m$	+0.016	[−.065, +.102]	+0.102	exploratory (development set)
Iterative, name-layer pool (IT-L0) − dense pool (IT-D)	OLD31 (31; questions)	answer correctness, $5{+}m$	−0.105	[−.234, +.011]	+0.011	exploratory (development set)
Event lookup lane on − off (not k-aligned)	GT v2 (500; questions)	answer coverage, 5 (+≤1)	+0.0089	[−.0030, +.0207]	+0.021	descriptive
Rebuild-2 − Rebuild-1 (release change: event registry and routing vocabulary; not a graph test)	GT v2 (500; questions)	verse recall@6	+0.0002	[−.0016, +.0026]	+0.0026	descriptive
(c) Legacy graph (Rounds 0–3 system), for context						
Round 3: graph − no-graph	Expansion (399; questions)	answer coverage@5	−0.014	[−.029, +.001]	+0.001	exploratory (not pre-registered)
Round 3: graph − no-graph	Expansion (399; questions)	anchor coverage@5	−0.012	[−.023, −.003]	−0.003[b]	exploratory (not pre-registered)
Round 3: graph − no-graph	All (499; questions)	hit rate@5	+0.002	[−.012, +.016]	+0.016	exploratory (not pre-registered)
Round 3: graph − no-graph	Original (100; questions; in-sample)	hit rate@5	+0.030	[−.020, +.080]	+0.080	exploratory (not pre-registered)
GN-41: KG bridge names − LLM rewrite	GN-41 (41; questions)	answer key score@5	−0.004	[−.035, +.026]	+0.026	exploratory (development set)
GN-41: KG facts to generator − LLM rewrite	GN-41 (41; questions)	answer key score@5	+0.018	[−.026, +.081]	+0.081	exploratory (development set)
(d) Non-graph context and ceilings						
LLM sub-query (QD) − deeper dense (DEEP)	NEW82 (82; clusters)	$f(q)$@6	+0.101	[+.058, +.147]	—	exploratory (development set)
QD − DEEP, step 2	GT v2 multi-anchor (204; questions)	anchor coverage@6	+0.0174	[−.0025, +.0380]	+0.038	pre-registered stopping rule
Context-conditioned sub-query (IT-D) − DK	OLD31 (31; questions)	answer correctness, $5{+}m$	+0.185	[+.054, +.331]	—[d]	exploratory (development set)
Gold bridge names (QD+ORACLE) − QD, rule O	NEW82 (82; clusters)	$f(q)$@6	−0.008	[−.027, +.008]	+0.008	pre-registered stopping rule
Gold passages (ORACLE) − DK	OLD31 (31; questions)	answer correctness, $5{+}m$	+0.347	[+.210, +.497]	—	exploratory (development set)
Author-written fact chain (FACTS) − DK	OLD31 (31; questions)	answer correctness, $5{+}m$	+0.503	[+.349, +.645]	—	exploratory (development set)

Rows are not comparable with one another. They differ in question set; in metric ($f(q)$ is retrieval hop coverage, answer correctness is graded by two LLM raters, the answer key score is deterministic, and GT v2 and Round 3 rows use the benchmark metrics); in k; in control; and in weighting (clusters or questions; see $\bar{\Delta}$ below). Read each upper bound only within its row. The GT v2 and Round 3 answer rows come from one generation per arm, and their intervals do not cover run-to-run drift. Intervals over questions ignore clustering; for OLD31, whose largest cluster holds 9 of its 31 questions, they may be too narrow, so its upper bounds may understate plausible gains. $\bar{\Delta}$: mean paired difference, first-named arm minus control. For "clusters" it is the mean of union-find cluster means (40 clusters on NEW82, 20 on OLD31) and the interval is a bootstrap over clusters; otherwise both are over questions. $f(q)$ is the fraction of a question's hops with a gold passage in the top-k; answer correctness is the mean over two Claude raters and three generations; both lie in $[0, 1]$. "$5{+}m$": the production top-5 plus m appended passages; DK always reads $5{+}m$, while KIN and L0 append fewer where they lack candidates, which favors DK. Intervals omit leading zeros; upper bounds are given where the interval includes zero or lies below it. No row is confirmatory; the "pre-registered stopping rule" rows on NEW82 ran on the development set. Rules G1, G3, O, and I are the pre-registered criteria of the October 18 run (Section XII-B). Round 3 rows use the legacy system and its own ground truth.

[a] Stopped by the pre-registered rule (≥ 3 of 20 clusters discordant and losses ≥ wins); this is not evidence of absence. Cluster W:L 2:2, 4 of 20 discordant. The scores are sealed by the pre-registration (no $\bar{\Delta}$ or CI). At the pre-registered discordance rate (0.222) the rule stops with probability 0.552 under the null and 0.072 at a cluster win rate of 0.8.

[b] Raw sign-flip $p = 0.013$; Holm-adjusted $p = 0.107$ over the eight tests {hit, verse recall, anchor coverage@5, MRR} × {all 499, expansion 399}.

[c] Not an upper bound and not evidence of a gain: GD was selected among about ten settings on NEW82 (cluster W:L 11:0, exact $p = 1/1024$); the pre-registered check on OLD31 (next row) stopped it.

[d] Uncorrected interval; Holm-adjusted $p = 0.123$. About 80% of the difference comes from the 18 questions that needed one appended passage.

TABLE XXI
MISSING HOPS REACHED BY THE CANDIDATE POOL AND KEPT AFTER CROSS-ENCODER SELECTION

	Graph arm		Control	
Arm	Pool	Kept	Pool	Kept
OLD31, 58 missing hops				
L0 lookup vs. DK	37	8	—	8
KIN walk vs. DK	17	6	—	8
Per-hop expansion[a]	—	21–22	37	21–22
NEW82, 102 missing hops				
GD vs. C, $k = 10$	59	31	35	17
GD vs. C, $k = 5$	59	14	35	7

Missing hops whose gold passage is in the candidate pool or among the kept passages. realkg keeps m passages per question. GD's pool includes the dense top-20 it shares with C, so its pool count is not the number of graph-proposed hops.

[a] Offline probe, 49 steps, IT-D's sub-queries fixed. Every graph-expanded pool keeps 21–22, as does a dense control of the same size, and gold passages brought in by the graph are kept 0 times. The control pool is IT-D's dense pool; perfect selection on it would keep 35.

21 PPR-brought bridge gold passages (14 hops), only the gold of a single hop outscored the best dense candidate of its step; their median score was 0.013 against 0.417. Feeding graph signals into selection made it worse: reciprocal-rank fusion of the reranker with PPR or with name matches kept 19 and 14 hops, against 21 for the reranker alone. On October 18, IT-GD's PPR candidates held a new-hop gold passage absent from IT-D's pool in 23 of 164 steps, but the reranker chose a PPR candidate 10 times, 2 of them hitting a new hop. This mirrors Round 3, where wrong graph anchors with surface overlap passed the same reranker (Section XI-C). HippoRAG 2 instead ranks passages by the PPR scores themselves [20]; we did not reproduce that design.

(B) No identity or relation layer, and low coverage. Rebuild-2 has neither layer, and its name keys merge homonyms. On the rebuilt text the kinship prototype (Genesis, Ruth, 1–2 Samuel, 1–2 Kings, 1 Chronicles 1–9, Matthew 1, Luke 3) gives two or more distinct fathers (or parents of unstated sex) to 163 of 622 children (26.2%), and 703 of its 801 cross-passage two-hop chains pass through such a node; the KIN pool reaches 17 of 58 missing hops. The legacy KG was no better: its `entity_id` (type plus toneless pinyin) produced 266 colliding ids, and its 15,926 semantic edges had a strict precision of about 0.23–0.28 (Appendix C). On the 84 GT v2 questions with a strong need for the KG it answered 25 completely, 44 partly, and 15 wrongly; it was wrong on 11 of 13 identity-disambiguation questions and on the only multi-hop question. On GN-41 it gave correct names for 9 of 104 answer-entity groups (3 strictly), 81 misses being missing data and 14 mechanism, and it produced any name for only 13 of 41 questions. This is the incomplete-knowledge regime in which KG-based RAG is known to degrade [10].

(C) Little graph-unique headroom, overlap with non-graph channels, and winner's curse. GD at $k = 10$ reached 14 missing hops that C missed (none the other way), but only about 6 of the 102 are reachable by the graph and by no non-graph channel (the script behind this count was not archived), and GD's lead over character-bigram BM25 and lexical pseudo-relevance feedback ($+0.052$, $+0.050$ at $k = 10$) would have failed its Holm step ($p = 0.0254 > 0.025$) even had the primary test been confirmed. Every apparent win over plain dense retrieval of a graph arm that includes an LLM rewrite came from the rewrite, which the non-graph twin reproduces: QD+G, QD+N, and QD beat DEEP by $+0.082$, $+0.099$, and $+0.101$; QD+N and QD, which differ only in the name line, differ by -0.002 $[-0.037, +0.038]$, an interval as wide as those of G1 and G3; and IT-D beat C7 by $+0.100$ $[+0.047, +0.155]$, to which PPR candidates added $+0.010$. GD's single-shot win at $k = 10$ involves no LLM; we attribute it to the small graph-unique headroom and to selection on the development set. GD was picked from about ten settings on NEW82, where 222 of 223 hops have a gold passage containing a body name (OLD31: 100 of 101); its gain depends on PPR's preference for name-dense passages (degree-normalized: $+0.006$ $[-0.030, +0.045]$ at $k = 10$); and 11:0 on NEW82 became 2:2 on OLD31, the winner's curse that the pre-registration had flagged in its power analysis. Other explanations — OLD31's different population, only four discordant clusters, a smaller true effect — are not excluded. The picture is consistent with graph benefit being task-dependent [25] and with GraphRAG gains shrinking under unbiased evaluation [11].

D. Closure and What It Does Not Claim

What was closed. Graph-retrieval variants on the current graph layer are closed. They were measured on the L0 name–passage graph: GD was stopped by V1, QD+G by rule G, and IT-GD, which rule I alone did not stop, was closed with them because the pre-run ruling tied the closure to rule G. The L1 event registry and the kinship surface triples, measured earlier (the event lookup lane on GT v2; KIN on OLD31), are closed by the authors' pre-run ruling. After the GD stop, no further single-shot graph variant was opened.

What this does not claim. The stops are not evidence of absence. Power is low: V1 rested on 4 discordant clusters of 20 and stops under the null with probability 0.552; the planned follow-up procedure, which would have rejected the null with probability 0.825 at a cluster win rate of 0.8, never ran; and realkg and iterq could detect only effects of 0.085–0.179. The data are development sets: NEW82 was seen and GD tuned on it, OLD31 is only partially blind, and routing and the event triggers were tuned on GT v2. The multi-hop sets are LLM-authored and unreviewed, and their answers were graded by LLM raters. The realkg and iterq intervals are over questions although OLD31 has a cluster of 9 questions, so they may be too narrow. GD and the October 18 arms were measured offline at the retrieval stage only. On October 18 only the L0 graph was exercised, and no layer with entity identity or reliable relations has been tested, so the result concerns this graph layer, not graph retrieval in general. The closure also covers designs we did not reproduce on this layer, notably ranking passages by PPR scores without the cross-encoder, as HippoRAG 2 does (mechanism (A)); it is a decision about further work, not a result about them.

Reopening condition. Reopening requires a new graph layer (e.g., entity identity, reliable relations) and a new pre-registration; under the attribution rule, a reopened test must again beat both a k-aligned dense arm and a non-graph twin.

XIII. Discussion

A. Three Co-Determinants of Observable Benefit

The central methodological claim of this paper is that in a GraphRAG system, **knowledge-graph construction quality, the retrieval ranking mechanism, and benchmark design jointly determine observable benefit** — and any one of the three can silently zero out investment in the others. The evidence chain is summarized by the original three-round chain (Rounds 0–2) on the legacy system: a completeness overhaul whose construction metrics all moved (entity anchor coverage $+14.4$ pt, event completeness $+50$ pt, cross-references $\times 273$) read as *zero* through a reranker-monopolized ranking onto a head-saturated benchmark (Section IX); one day of ranking-layer repair then released more measured value than the week-scale re-extraction program was projected to deliver on the same benchmark (Section X) — an in-sample comparison, since that repair was developed on the same 100 questions. The practical rule we draw: *optimization order should follow the bottleneck layer, not the upstream-first order of the data flow* — and the bottleneck is found by per-question forensics, not by metric staring.

Round 3 adds the converse. The same co-determinants can also silently *inflate* a reading: on a head benchmark whose failing graph anchors had been repaired question by question, the graph was credited with part of a margin that a no-graph control assigns mostly to signal routing ($+0.13$ of $+0.16$ on the same 100 questions), while on the 399 new questions no graph gain is detected (answer-coverage upper bound $+0.001$; Section XI). Benchmark design therefore decides not only whether a layer's investment is visible but also to which layer it is credited, and the instrument that settled the attribution was an ablation removing one layer while holding the others fixed.

The rebuilt system bounds the claim (Section XII). The co-determinant frame explains the legacy results, whose positive graph readings were in-sample, but it does not imply that a graph holds value which fixing one co-determinant will release. On the rebuilt layer the benchmark co-determinant was addressed directly — targeted multi-hop question sets, k-aligned dense controls, non-graph twins, an attribution rule fixed before the October 18 runs (and applied retrospectively to the earlier ones), and pre-registered stopping rules — and still no gain was detected. Giving PPR-ranked names to an LLM decomposer had an upper bound of $+0.023$ in hop coverage (a mean over question clusters) against its non-graph twin, and adding PPR candidates to iterative retrieval an upper bound of $+0.027$ (an interval resting on the two untied clusters of 40; both under a pre-registered stopping rule); dense-seeded PPR alone (GD) had an upper bound of $+0.054$ at the production $k = 5$ (exploratory, development set). The three failure mechanisms of Section XII-C touch all three co-determinants at once. (A) Selection (ranking): the cross-encoder scores passages only against the question or sub-query, so, in the per-hop and appended-passage settings, graph-proposed bridge passages reach the pool but rarely survive to the top-k. (B) No identity or relation layer (construction): the rebuilt layer has no entity-identity or relation layer, so names collapse homonyms, and on the GN-41 questions the legacy graph supplied only 9 of 104 needed answer-entity groups (3 strictly). (C) Headroom and evaluation: on NEW82 only about 6 of 102 missing hops appear reachable by the graph and by no non-graph channel (a count whose script was not archived), the apparent gains of graph arms with an LLM decomposer over plain dense retrieval were reproduced by their non-graph twins, and the one positive setting, chosen on the seen development set, fell from 11:0 to 2:2 on a second set. Within the limited power of these sets (Section XIV), the targeted yardstick did not reveal graph value that the 500-question benchmark had hidden.

TABLE XXII
Four Incidents: Graph Value Gated — or Wrong Graph Evidence Passed — by Retrieval-Side Machinery

Incident	Asset in the graph	Gating mechanism	Resolution
Round 0 (May)	entity vectors (EQ) bridging modern phrasing	reranker surface-form monopoly	EQ disabled; re-enabled under fusion
Round 1 (P0)	correct anchors newly in pool; TSK breadth	same monopoly + vote-ranked topical noise	fusion + provenance-stratified weights
Round 2 residual: dual-anchor family	*all* needed cross-book edges present	1-hop expansion depends on seed correctness	open; non-graph query decomposition tested Oct. 2026, below its pre-registered bar (§XII-B)
Round 3 (Sept.)	entity and event anchors, now fed by a working classifier	wrong anchors enter the pool; neither reranker nor fixed prior demotes them	graph strategies retired in the rebuild; graph operations tested Oct. 2026: no gain detected (§XII)

Rounds 0–3 ran on the legacy system; resolutions as of October 2026. The rebuilt-system experiments are not a fifth incident of the same kind: in the per-hop and appended-passage settings, the bridge passages their graph proposed were also rarely selected, but few missing hops were reachable only through the graph, so selection was not the only limit (§XII-C).

B. Four Incidents of the Same Lesson

Table XXII aligns the four incidents. The third deserves elaboration because it was discovered *after* the fusion remediation and is therefore an independent probe. The two GENERAL misses remaining after Round 2 are one family: cross-book *prophecy–fulfillment* questions requiring two anchor sets simultaneously (GENERAL_006: Jeremiah's new covenant in Hebrews, gold = Jer. 31:31–34 + Heb. 8; GENERAL_008:

Zechariah's prophecies in Passion Week, gold = Zech. 9:9; 11:12–13; 12:10 + Matt. 21/27). Live verification of the legacy graph showed that **every needed cross-reference edge existed** — Jer. 31↔Heb. 8 (3 edges), Zech. 9↔Matt. 21 (2), Zech. 11↔Matt. 27 (5), Zech. 12↔John 19 (2); TSK is *designed* for exactly this relation. The failure was upstream of ranking: retrieval seeds landed in the wrong chapters ("prophecy" surface forms hit the false-prophet conflict of Jer. 26/28/29; "Zechariah" hit the Zerubbabel visions of Zech. 4), and 1-hop expansion from wrong seeds could not reach right edges. Assets in the graph; value gated by the seed-selection mechanism. These two questions also posted the two lowest answer-correctness scores of the original suite (≈ 0.13, stable across re-judged runs). Round 3 populated the family (prophecy_fulfillment, $n = 26$), and its reading is consistent with the diagnosis: with the graph on, hit rate is 0.92 and anchor coverage@5 0.65, against 0.96 and 0.68 with it off — expansion from imperfect seeds adds nothing measurable (edge presence and seed placement were not re-verified for the new questions). In Round 3, GENERAL_008 itself hit, through the entity traversal that the repaired classifier activated rather than through cross-references. The planned fix, query decomposition, was later tested on the rebuilt system in non-graph form, since the rebuilt knowledge layer has no relation layer and hence no cross-reference edges to join on (Section XII-A). On the 204 multi-anchor questions of GT v2 (the 500 benchmark questions as evaluated on Rebuild-2), appending the best passage for one LLM-written sub-query to the same five passages changed anchor coverage@6 by $+0.0174$ (95% CI $[-0.0025, +0.0380]$; question W:L 29:20) relative to appending a passage from a deeper dense pool. That missed the pre-registered bar (at least $+0.03$ with a CI lower bound above 0), so the multi-anchor line was stopped by the pre-registered rule; this is not evidence of absence (Section XII-B). The dual-anchor family remains open.

The fourth incident inverted the first two. There, the ranking layer discarded correct graph evidence; in Round 3 it passed graph evidence whether or not it was correct, because neither of its signals concerned the anchor: the reranker scored surface form and the prior scored the proposing strategy. On the 29 questions where graph strategies took all five context slots, the graph run trailed the no-graph run by 0.10 verse recall and 0.11 coverage (a descriptive split on the graph run's own output), and in most graph-caused losses the reranker alone already preferred the injected passages (Section XI-C). Injecting graph candidates is a safe default only for verified anchors; fusion compounds the problem rather than causing it. The rebuild took the conservative course: apart from routing vocabulary drawn from its name layer, its only online knowledge-layer signal is a curated event lookup lane that appends at most one curated anchor after the top-k instead of injecting candidates into the pool. Even this lookup, which is not a graph operation, showed a retrieval gain only in sample, on the 23 of 500 questions where it fired, and switching it on versus off moved answer coverage by $+0.0089$ (95% CI $[-0.0030, +0.0207]$; not k-aligned; descriptive; Section XII-B).

C. Retrieval-to-Answer Attenuation

Post-fusion, context recall rose $+0.053$ and answer coverage $+0.041$, while answer correctness moved only $+0.021$ — at the edge of judge noise. EVENT_005 illustrates both limits of reading such numbers. On the unit-level metrics its retrieval is perfect (context recall 1.0, recall@5 1.0), yet its answer scored 0.4 coverage and 0.173 correctness on the run of record, and re-judging the identical outputs moved those to $0.70/0.353$. The verse-level metrics introduced later show that the retrieval was not perfect: the gold is Numbers 13–14 and no archived run retrieved chapter 13, so on the run of record anchor coverage@5 is 0.5 and verse recall 0.29. And in Round 3 the no-graph and semantic-only runs received identical contexts for it yet scored 0.7 and 0.4 coverage. Faithfulness is flat across all three time points (≈ 0.95) and, under the repaired judge, ≈ 0.98 in every Round 3 configuration, which is consistent with the gains coming from better context; what is lost is shared among retrieval breadth that unit-level recall hides, the small generation model (Gemma 4 E4B), and the judged metrics themselves. The quality bottleneck is therefore no longer retrieval alone — consistent with our framing that the observable-benefit chain must be examined end to end. Any answer-model A/B here must first neutralize the noncommittal-refusal artifact of Section VI-B, or the honest model loses by construction — the local instance of the evaluation-bias warnings in [11].

D. Benchmark Saturation: Evaluation Design as Prerequisite

At 97/100 (VERSE, TOPIC, EVENT at 1.00 hit rate), the benchmark is saturated in the head, and the three residual misses belong to two families that need *new architecture*, not better ranking: cross-chapter aggregation (PERSON_004; the community-summary/RAPTOR layer [3], [22]) and cross-book dual-anchor conjunction (GENERAL_006/008; query decomposition). (Round 3 qualifies both: without the graph PERSON_004 hits at the unit level, its graph-mode miss being the displacement of Section XI-C, although complete coverage still needs aggregation; and GENERAL_008 then hit through the classifier-activated entity traversal.) Consequently, *every* roadmap item then remaining — long-tail re-extraction (P1), event-temporal modeling (P2, [23], [24], [34]), hierarchical aggregation (P3) — would read as zero on this yardstick: P1's tail improvements are unread by head questions (consistent with P0, although Round 3's long-tail families show no graph benefit either; Section XI), P2's target questions already retrieved perfectly at the unit level (their verse recall is only 0.43; see below), and P3's target population is $n = 1$. The methodological consequence: **benchmark redesign is the prerequisite for all further optimization**, not a reporting afterthought. The needed instruments have since been built (below and Section XII-A); the general principle — that the benchmark itself gates which layer's investment is visible — echoes the defect-sensitivity methodology of BRINK [10] and the task-dependence findings of [25]. A cautionary corollary from Section X: with $n = 2$ family sizes, tuning the system on the residual misses would be overfitting; the family must be *populated first*, then fixed.

That prescription has since been carried out. The 500-question benchmark (Section VI-A) populates the aggregation and dual-anchor families (trajectory $n = 18$, multi_chapter $n = 25$, prophecy_fulfillment $n = 26$, among 19 families), and its first reading, with the no-graph control, revised the graph's share of the margin and showed that the graph's head gains do not transfer to new questions (Section XI). It also corrects one premise above: P2's target questions do *not* retrieve perfectly once breadth is measured — the temporal family reads verse recall 0.42 and anchor coverage@5 0.53 under the graph configuration — and the aggregation families sit lower still (verse recall 0.34 for trajectory, 0.37 for multi_chapter). The yardstick has room again for multi-anchor retrieval in general, though not enough to detect a graph gain: only one of the 500 questions has a multi-hop relation as its primary type and only 15 need two or more hops, and even a perfect repair of every non-curated question the legacy graph could serve would have raised verse recall by about $+0.018$ (at most $+0.030$), 1.2–2 times the minimum detectable effect. The 500 questions can reliably detect graph harm, not a graph gain (Section XII-A).

For the graph, the prescription was then carried out in October 2026, mostly on the rebuilt system. Three targeted multi-hop sets were built — NEW82 (82 questions, 223 hops), OLD31 (31 questions, 101 hops), and GN-41 (41 questions on the legacy graph) — and each graph arm was compared with a k-aligned dense control and, where an LLM rewrote the query, with a non-graph twin that differed only in the graph operation (Section XII-A). No graph arm showed a detectable gain (Table XX) except in one development-set reading: dense-seeded PPR at $k = 10$ (cluster W:L 11:0, selected among about ten settings; at the production $k = 5$ its upper bound is $+0.054$). In the pre-registered direction check on OLD31 it fell to 2:2 (4 of 20 clusters discordant) and was stopped by the pre-registered rule; this is not evidence of absence. On the current layer, then, a yardstick built to see graph value detected none; with these set sizes that argues against a large hidden gain but cannot exclude a small one. The principle stands — a yardstick must be able to see a layer before that layer's investment is judged — but a yardstick that can see a layer may simply report that the layer adds nothing detectable.

E. Patch Lifecycles Under Architecture Evolution

The pin audit (Section V-F) generalizes: the EQ pin and graph-uncertainty pin were correct *for the architecture they were built in* — discrete rescues under a ranking that discarded graph signal. Once fusion transmitted that signal continuously, the same pins became net-negative (noise promotion), and the "regression" observed when retiring them (GENERAL NDCG -0.08) was itself an artifact of the artificial placements they used to manufacture. Remediation machinery must be re-audited whenever the layer it compensates for changes; yesterday's fix is today's bug. We suspect this lifecycle is common in deployed RAG systems, where discrete patches accrete around ranking layers precisely because ranking is the last mile.

F. Cost and Privacy

All legacy-round conclusions above were reached with every model of the system local — embedder, reranker, extraction classifiers, generator, and judge — on a single GPU host; the rebuilt-system experiments kept the retrieval stack and the decomposer local, but their multi-hop questions were written, and the OLD31 answers graded, by a commercial LLM (Claude; Section XII-A). Round 0 established that a locally served 8-bit E4B-class model matches a commercial API model end to end on this task within 4 pp on six of seven metrics (Table VIII). Round 3 qualifies but does not overturn that result — Gemma's Round 0 classifier was inactive, and its faithfulness values predate the judge repair — and an answer-model A/B under the repaired instruments remains to be run; all optimization value so far has been retrieval-side. For sensitive religious-domain deployments — where both the corpus and user questions may be confidential — this is the operative cost/privacy result: the investment frontier is retrieval and evaluation engineering, not model procurement.

XIV. Limitations

Rebuilt knowledge layer. The rebuilt system has no entity-identity or relation layer: its name layer (2,926 names) puts homonyms on one node, and its 31-event registry fires on literal triggers. On unseen event questions the registry fired on too few questions (13 of 61) for its pre-registered test, with precision 11/13 and recall 11/43 (descriptive), and 15 of the 23 benchmark questions on which it fired matched external aliases tuned on the same ground truth. Of the legacy graph that Rounds 0–3 ran on, the residuals listed in earlier versions (incomplete pericope-level MENTIONS, alias coverage of $\sim$0.8% of entities, unclassified relation candidates, no coreference resolution, and toneless-pinyin IDs that over-merge homophones, 266 of them colliding) no longer bear on the deployed system, which retired that graph.

Rebuilt graph experiments. The graph experiments of Section XII, none of which is confirmatory, carry six further limits. (i) *Development sets.* NEW82 is a seen development set: the dense-seeded PPR configuration was chosen among about ten settings on it, and the name formula of the decomposer arms was revised after inspecting the name lists of six of its questions. OLD31 is only partially blind: 16 of its 31 questions share gold passages with NEW82, and the PPR seed type was chosen after its pool-level results were seen. (ii) *LLM-authored, unreviewed sets.* NEW82, OLD31, and GN-41 were authored by an LLM (Claude) and never reviewed by a human, and the OLD31 answers were graded by two Claude raters, whose agreement (linear κ 0.98) measures reliability, not validity; three OLD31 questions score zero in every arm, including the arm given the gold facts, which suggests faulty items. (iii) *Low power.* The pre-registered direction check that stopped dense-seeded PPR would have stopped with probability 0.552 under the null and 0.072 at a cluster win rate of 0.8, so its 2:2 argues against a cluster win rate as high as 0.8 but says little about moderate or small effects; the OLD31 answer-level comparisons have minimum detectable effects of 0.085–0.179 at 80% power; and the 500-question benchmark can detect graph harm but hardly a graph

gain. (iv) *One generation per arm.* Round 3 and the GT v2 comparison of the event lane generated each answer once, and their question-resampling intervals do not cover run-level drift (a Rebuild-1 A/A rerun moved answer coverage by $+0.0149$); the three decompositions averaged by the decomposer arms are not independent samples. (v) *Question-level intervals.* The GD and decomposer-arm intervals resample clusters of questions that share gold passages, but the OLD31 answer-level, GN-41, GT v2, and Round 3 intervals resample questions; OLD31's largest cluster holds 9 of its 31 questions, so its intervals may be too narrow. (vi) *Selection and instruments.* No family-wise control spans the rebuilt-system program, which ran well over a dozen comparisons; its positive exploratory readings (GD at $k = 10$, QD on NEW82) carry that multiplicity as well as their selection on the development set. Mechanism (A) was observed with one cross-encoder (BGE-reranker-v2-m3, 512-token input) and the sub-query arms with one small local LLM (Gemma 4 E4B), so other selectors or decomposers may behave differently. The count of about 6 graph-only missing hops behind mechanism (C) cannot be reproduced, because its script was not archived. Finally, GD and the October 18 arms were measured at an offline retrieval stage (host-side exact dense search, without routing or the event lane) and at the retrieval level only.

Benchmark scale and design. Rounds 0–2 rest on 100 head-entity questions, on which the Round 2 repairs were also developed; per-category deltas below ≈ 0.05 on judged metrics are noise there, and several of their architectural claims rest on question families of size 1–2, which we deliberately refused to tune against. Round 3 adds 400 questions in 19 families, but the families are small (8–48 questions), each configuration was run once, and the ground truth has known weaknesses: 313 of the 500 items carry exactly five expected points (a mechanical decomposition of the gold passage), and some points are broader than what the question asks, so coverage levels are conservative — comparisons between configurations, which share the ground truth, are less affected. The 400 new items were content-checked in an LLM-assisted review rather than by the annotators of the original 100. The expansion set is diagnostic rather than representative: its families target known failure modes, some of which favor dense retrieval and some of which expose graph gaps, so the Round 3 reading of the graph — no gain detected, with an upper bound of $+0.001$ on the expansion answer-coverage difference — describes this mix, from one run per configuration. Its harm direction is consistent, but the mean-difference harm does not survive a Holm adjustment [41] over the eight retrieval tests of the later audit (expansion anchor coverage@5, -0.012: raw p 0.013, adjusted 0.107; a retrieval-only rerun gives p 0.060). Only the first-gold-rank win:loss count survives a Holm adjustment applied to the sign tests of the same eight cells (12:35 on the 399 expansion questions, sign test p 0.001), and it replicated in the rerun (13:32, p 0.007); that statistic was chosen after the results had been seen. Finally, the semantic-only baseline has no reference parser, so most of the routing gain is the use of a reference that the user supplied; and the book-anchor change deployed in all Round 3 configurations was itself checked on the 500 questions.

Judged metrics. RAGAS scores carry ± 0.4 per-question noise at our scale and a noncommittal rule that penalizes honest refusal; the answer-coverage judge is a single local model, and on byte-identical contexts coverage still moves by 0.060 per question on average from generation sampling plus judging (Section XI). Faithfulness values before the September 2026 judge repair are systematically understated. All retrieval claims are therefore anchored to deterministic metrics.

Configuration history. In the Gemma configurations of Rounds 0–2 the intent classifier was silently inactive, so routing was dictionary-only and the classifier-fed strategies never ran. Round 3 is the first local-model evaluation with a functioning classifier (Claude's worked in Round 0). The two regimes meet on the original 100 questions (graph hit rate 0.97 in both) and on a July 15 run of all 500 questions (graph-configuration verse recall 0.751 vs. 0.749), but the graph/no-graph contrast exists only under the repaired classifier, whose re-routing (e.g., 48 fallback→R4) increases the graph's exposure. All four rounds also ran dense-only, with a query encoder inconsistent with the index, on a lossy legacy conversion of the corpus (Appendix C).

Generation. The deployed generator is an 8-bit E4B-class model; the attenuation analysis shows it discards part of the retrieval gains. Conclusions about answer quality are conservative lower bounds.

Scope. One corpus, one language, one translation, one domain. The architecture's signal set and dictionaries are domain-engineered; transfer requires re-instantiating those assets (though the methodological findings — ranking-layer gating, benchmark dependence, patch lifecycles — are domain-agnostic claims about system structure, not about the Bible). The rebuilt-system null concerns one name-level knowledge layer over one corpus; it says nothing about graphs that have identity and relation layers.

XV. Future Work

Evaluation first. Per Section XIII, the next artifact was the benchmark, not the pipeline. Done since the first version of this paper: long-tail question sets, populated cross-chapter-aggregation and dual-anchor families (Section VI-A), the faithfulness repair, and, for the graph, targeted multi-hop sets with k-aligned non-graph controls and pre-registered stopping rules (Section XII-A). BRINK-style defect injection [10] — remove known edges, measure answer degradation — was listed here as a causal probe of KG value; it presupposes a measurable graph contribution to remove, and none was detected on the current layer, so it is deferred until a layer shows a detectable contribution. Still open: (i) noncommittal exclusion and position/length debiasing [11]; (ii) an audit of the expected answer points; (iii) repeated runs per configuration; and (iv) human review of the LLM-authored multi-hop sets.

What reopening the graph line would require. Graph retrieval on the current layer is closed (Section XII-D). It was measured on the L0 name–passage graph (dense-seeded PPR, GD; PPR-ranked name hints to an LLM decomposer, QD+G; PPR candidates in iterative retrieval, IT-GD): GD was stopped by its pre-registered direction check and QD+G by rule G,

and IT-GD, which its own rule did not stop, was closed with them because the authors' pre-run ruling tied the closure to the failure of rule G; this is not evidence of absence. The L1 event registry and kinship surface triples, measured earlier (the event lookup lane; the kinship walk KIN on OLD31), are closed by the authors' pre-run ruling. Reopening requires a new graph layer (e.g., entity identity, reliable relations) and a new pre-registration. In the authors' judgement, a reopened line would need five things, matched to the mechanisms of Section XII-C. (1) *An entity-identity layer*, so that homonyms stop sharing a node. The decoupled re-extraction once planned here as P1 is the obvious route — whole pericopes as the extraction unit (storage unit $\neq$ extraction unit [40]), a coreference pass following LINK-KG [9], true-loop gleanings [3], [42], and post-hoc entity resolution with propose-then-confirm merging — but it is no longer planned work: the rebuild stopped the knowledge layer at a name layer and an event registry. (2) *Reliable relations.* The legacy semantic edges had a strict precision of only about 0.23–0.28, so a relation layer would need ontology-grounded correction [28], [29], cross-chunk assembly [31], [33], and denoising checked downstream [30]; the event-temporal layer once planned as P2 [23], [24], [34] is likewise not planned. (3) *A blind, held-out, human-reviewed multi-hop question set*, written before any arm is designed or tuned. (4) *Selection that can score bridge passages*: a selector that judges a passage by the hop it bridges, not only by its match to the question or sub-query — for example PPR-ranked passages as in HippoRAG 2 [20] or calibrated graph–vector fusion [18]. Plain reciprocal-rank fusion of reranker and PPR rankings did not help in our one test: in a per-hop probe on OLD31 it selected 19 of 58 missing hops, against 21 for the reranker alone (descriptive). (5) *The attribution protocol* of Section XII-A: only PPR or a traversal of two or more hops counts as a graph operation, LLM query rewriting does not, and a graph arm must beat both a k-aligned dense arm and its non-graph twin.

Aggregation retrieval. Community summaries (Leiden partitions [43] as in [3]), hierarchical community retrieval [21], or RAPTOR-style summary trees [22] for cross-chapter aggregation remain untested. They are not retrieval variants on the closed graph layer, so the closure does not cover them, but they should be judged by the same protocol. The multi-anchor questions they target remain open: the non-graph query decomposition aimed at them missed its pre-registered bar (Section XII-B).

Retrieval mechanics. The retrieval repairs listed here in earlier versions are no longer open. Anchor-confidence gating, per-strategy slot quotas, and the $\alpha = 0$ run that would have separated injection from prior effects all concerned graph candidates in the ranked pool, which the rebuild removed; query decomposition and PPR candidates on the name layer were tested on the rebuilt system (Section XII), and graph-native scoring on that layer — ranking passages by PPR scores without the cross-encoder, as HippoRAG 2 does [20] — falls under the closure although we did not reproduce that design on this layer; the closure is a decision about further work, not a result about that design. The only item those experiments left open is non-graph: a blind confirmation of LLM sub-query rewriting (QD). On the NEW82 development set, QD raised hop coverage (cluster-averaged) over a passage from a deeper dense pool by $+0.101$ (95% CI $[+0.058, +0.147]$; cluster W:L 19:2; exploratory, development set); on the 204 multi-anchor questions of GT v2 it missed its pre-registered bar ($+0.0174$, $[-0.0025, +0.0380]$). The blind confirmation was not run, by decision.

Generation. Answer-model A/B under debiased judging, since the attenuation analysis locates a further bottleneck there.

XVI. Conclusion

We presented the complete architecture of the GraphRAG system that served Traditional Chinese Bible QA in Rounds 0–3 — from layout-geometric PDF parsing, pericope-centric chunking, grounded dual-track entity extraction, closed-schema relation extraction, and a three-source cross-reference network, through a signal-adaptive six-route retrieval design terminated by cross-encoder reranking and linear rank fusion — and an unusually complete empirical record of optimizing it: a quantified audit tracing KG damage to granularity inheritance rather than chunk parameters; a completeness overhaul whose construction metrics all moved and whose benchmark reading was zero; the question-level diagnosis of that negative result; and a one-day ranking-layer remediation that surpassed all prior results (hit rate 0.97, EVENT 1.00, $+0.16$ over the semantic baseline), decomposed by ablation into complementary discrete and continuous mechanisms. A fourth round, on a 500-question benchmark with repaired instruments and a no-graph control, replicated the head hit rate and margin and re-attributed them: about four-fifths of the margin over semantic-only belonged to signal routing, the graph's share ($+0.03$, not distinguishable from zero; upper bound $+0.08$) was concentrated in head events with hand-curated anchors, and on 399 new questions no graph gain was detected (verse recall@5 -0.009 and answer coverage -0.014, each with upper bound $+0.001$, from one run per configuration), with wrongly anchored graph candidates passing a ranking layer that could not tell them from right ones; the harm direction survived a multiplicity adjustment only in a first-gold-rank count chosen after the results were seen. The same round uncovered an intent classifier that had been silently inactive in every earlier local-model run.

The rebuilt system, whose knowledge layer is reduced to a name layer and a curated event registry, then gave the graph a targeted test, and graph retrieval showed no detectable gain outside the development set: against k-aligned dense controls and, for the decomposer arms, non-graph twins on LLM-authored multi-hop sets, the upper bounds on the hop-coverage difference (averaged over clusters of questions that share gold passages) were $+0.054$ for dense-seeded PPR at $k = 5$ (exploratory, development set), and $+0.023$ for PPR-ranked name hints to an LLM decomposer and $+0.027$ for PPR candidates in iterative retrieval (pre-registered stopping rules), while the apparent gains of the decomposer arms over plain dense retrieval were reproduced by their non-graph twins. On the L0 name–passage graph, dense-seeded PPR and the PPR name hints were stopped by their pre-registered rules.

PPR candidates in iterative retrieval (IT-GD), which their own rule did not stop ($+0.010$, upper bound $+0.027$), were closed with them by the authors' pre-run ruling. This is not evidence of absence. The L1 event registry and kinship surface triples, measured earlier, are closed by the authors' pre-run ruling, and reopening requires a new graph layer (e.g., entity identity, reliable relations) and a new pre-registration.

Three conclusions, we believe, carry to other GraphRAG deployments. First, the measurable gains were not the graph's: signal routing carried about four-fifths of the Round 3 margin, and on the rebuilt system the pipeline's gain over semantic-only ($+0.032$ answer coverage on GT v2, a development set; descriptive; Appendix C) was concentrated in questions that cite a reference explicitly ($+0.092$ on 179; -0.002, upper bound $+0.017$, on the other 321). Second, graph gains were in-sample: the legacy graph's share appeared on head questions whose anchors had been built or verified on those very questions and did not transfer to new ones, and on the rebuilt layer no route of transmission — the reranker selecting PPR candidates (apart from a post-hoc $k = 10$ reading on NEW82 that fell to 2:2 on OLD31), PPR names given to an LLM decomposer, or even gold bridge names (upper bound $+0.008$ over all 82 questions, only 24 of which had eligible names) — produced a detectable gain; on NEW82 dense-seeded PPR at $k = 10$ reached only 14 of 102 missing hops that its dense control missed, and, by a count whose script was not archived, only about 6 were out of reach of every non-graph channel, so there appears to have been little to transmit. Third, the methods that settled these readings are portable: establish a layer's contribution by ablation on questions its repairs never saw; compare a graph arm with a k-aligned dense arm and with a non-graph twin that differs only in the graph operation; and fix the stopping rule before the run. The structural observations of the legacy rounds stand alongside: a ranking layer can hide correct graph evidence (Rounds 1–2) or pass wrong graph evidence (Round 3); benchmark design decides which layer's investment is visible, and to which layer it is credited; discrete patches have architecture-bound lifetimes; and optimization order should follow the measured bottleneck, not the data flow. In our deployment, the bottleneck rule delivered, in a day of work, a larger in-sample return on the head benchmark than the deferred week-scale re-extraction plan was projected to yield there; the ablation rule then showed that the graph's share of that return did not transfer beyond the questions it was built on; and the rebuilt-system experiments found no detectable graph share to transmit on a name-level layer.

APPENDIX A
STAGE-TO-CITATION MAP

Table XXIII indexes, for every pipeline stage and experimental round, the literature it draws on and the role each work plays — so that each citation can be audited in place.

APPENDIX B
REPRODUCIBILITY NOTES

Determinism and re-scoring. All retrieval metrics are computed programmatically against the gold references (at unit, verse, and chapter level); every evaluation run archives per-question raw responses (sources, strategies, raw reranker and fused scores, route), and the metric suite can be recomputed bit-exactly from the archives. The fast retrieval-only loop was validated to agree bit-exactly with the full pipeline on all retrieval metrics in Rounds 0–2, while the intent classifier was inactive. Retrieval is deterministic *given the classifier output*; routes and graph inputs depend on the LLM intent classifier, which samples at temperature 0.1 — in Round 3, 12 of 500 questions (2.4%) were routed differently in two configurations whose routing logic is identical. The script `paper/tools/round3_stats.py` recomputes every number derived from the three Round 3 archives (aggregates, bootstrap intervals with a fixed seed, the noise floor, the slot analysis, the faithfulness gate, the traces) after asserting that its aggregates equal the archived ones; `paper/tools/replay_routes.py` reproduces the empty-classifier route replay and the July 15 comparisons from the git-archived runs, but it imports backend modules that the October 2026 rebuild removed, so it must be run from a worktree checked out at a legacy commit.

Comparability across the three time points. Answer model (Gemma 4 E4B, 8-bit), judge model (Gemma 4 26B), decoding temperature (0.1), and top-k (5) were verified identical, as was the route distribution (R1=20, R2=25, R3=21, R4=18, R5=10, R6=4, fallback=2) at the first two time points; at the third, one question (EVENT_015) moved from R6 to R4 after the event keyword "Gethsemane prayer" was added in the fusion round. All three time points ran with the intent classifier silently inactive (Section V-C): replaying the signal detector with an empty classifier output reproduces every logged route. The entity-query strategy was off at the first two time points and on after the fusion round (its re-enablement is itself one of the audited repairs). The only other systematic difference between time points is the state of the knowledge graph and of the post-retrieval chain, as described in the text.

Comparability with Round 3. Round 3 differs from the three time points in the classifier (functioning), book-anchor retrieval (per book, developed and checked on the 500 questions), merged-verse lookup, the judges' inputs (generator context blocks) and faithfulness rubrics, the Ollama server version, and the benchmark (500 questions). The bridge is the original 100 questions scored by the deterministic retrieval metrics, which can be recomputed from every archive; across it the graph configuration's hit rate (0.97) and its margin over semantic-only ($+0.16$) are unchanged.

Rollbackability. While the legacy system was deployed, every P0 and fusion-round mutation was reversible: backfilled mention edges, co-occurrence relation edges, and TSK cross-reference edges carried provenance flags and could be deleted by a single predicate each; curated event nodes carried a source tag; alias injections had before-snapshots; and the rank-fusion layer, expansion caps, and entity-query strategy were configuration switches whose legacy settings reproduced the pre-remediation system exactly. Neo4j and the legacy data were deleted in the October 2026 rebuild (Appendix C), so the legacy graph and service can now be restored only from the pre-rebuild backup.

TABLE XXIII
WHICH WORK IS CITED AT WHICH STAGE, AND WHY

Stage (section)	Works	Role of the citation
Problem framing (§I)	[1]–[4], [12]	faith-dimension LLM deficits motivating grounded QA; RAG foundations and surveys; GraphRAG paradigm
Chunking & granularity (§III-D, §VIII M5–M6)	[3], [16], [27], [34]	tokenizer/embedder fit; chunk size → extraction density (600 vs. 2,400 tokens ≈ 2× entity references); finer units extract more completely; long-context degradation as the upper bound
Coreference & overlap (§VIII M4)	[6]	40.7% of relational facts cross-sentence, 17.6% need coreference — the structural ceiling of fragment-local extraction without coreference (used as *indirect* evidence; no peer-reviewed overlap-size quantification exists)
Entity extraction (§IV-A)	[7], [8], [17], [37]	Chinese segmentation/NER; the local classify-only LLM; structured prompting as noise suppression
Relation extraction (§IV-B)	[26], [32]	schema-based paradigm positioning; evidence-span grounding against context-divorced triples
Cross-references / TSK (§IV-C)	[5], [30]	cross-reference data (OpenBible.info, CC BY); repair-by-reweighting rather than deletion
KG-quality remediation roadmap beyond P0, not executed (§VIII, §XV)	[9], [28], [29], [31], [33], [40], [42]	coreference recipe (-45% duplication, length-amplified); post-hoc canonicalization / correction; cross-chunk reassembly; storage $\neq$ extraction unit; true-loop gleanings
Retrieval substrate (§V-E)	[13]–[16]	dense retrieval; BM25 and RRF hybrid (designed, never executed online); embedder and cross-encoder
Rank fusion (§V-F, §X-C)	[18]	calibrated graph/vector fusion as the sophisticated alternative to our single-α linear form
Evaluation framework (§VI)	[11], [35]	RAGAS metrics; judge-bias methodology (position/verbosity; noncommittal artifact)
Negative result & benchmark dependence (§IX, §XI, §XIII)	[10], [11], [25], [41]	KG-defect → answer non-linearity and defect injection; debiased gains; when graphs help RAG at all; multiplicity adjustment of the Round 3 graph − no-graph tests
Graph retrieval on the rebuilt system (§XII)	[10], [11], [20], [25], [41]	personalized PageRank as the operator family of the PPR arms (GD, QD+G, IT-GD; not a reproduction of HippoRAG 2, which ranks passages by their PPR scores); Holm adjustment where a test family was declared; degradation under incomplete knowledge (mechanism (B)); task-dependent graph benefit and debiased GraphRAG gains (mechanism (C))
Aggregation, event-temporal layer & graph-native selection (proposed, not built) (§IX-D, §XV)	[3], [18], [20]–[24], [34], [43]	RAPTOR trees / Leiden and hierarchical community summaries for cross-chapter aggregation (untested); temporal–causal event modeling (the event-temporal layer, no longer planned); passage ranking by PPR scores as a selector for bridge passages, as in HippoRAG 2, which the rebuilt-system experiments did not reproduce, or calibrated graph–vector fusion
Retrieval-side chunking sensitivity (§II)	[36]	cited *only* for retrieval-downstream chunking sensitivity — explicitly not extrapolated to KG construction

Rebuilt-system graph experiments. The new graph experiments of Section XII (GD, realkg, iterq, the per-hop probes, and the October 18 run) ran offline on the data layer of Rebuild-2; GN-41 ran on the legacy KG before the rebuild, and the event-lane and release comparisons on the served builds. The pre-registrations and code of GD and the October 18 run are in the repository: `evaluation/experiments/2026-10-14_graph_gd/` (`prereg.md`; `evaluate.py`, whose subcommands recompute `new82_report.json`, `v1_old31.json`, and `units_old31.json`; `v1_power.py`, which simulates the stop rule's operating characteristics) and `evaluation/experiments/2026-10-18_graph_qd/` (`plan.md`, committed in 16f9ced before the run started and updated with the results in 533b4ee; `qd_eval.py`; `qd_step2.py`). The archived outputs are kept outside the repository, in the project's artifact store under `reports/`: `graph_gd_20261014/`; `graph_qd_20261018/` (`report.json`, `step2/report2.json`); `kg_effect_eval_20261014/new82_ans/` (`retrieval_iter.json`, the iterative arms); `multihop_kg_20261012/` (the `realkg` and `iterq` analyses on OLD31, each with its `analyze.py`, and the per-hop probes); `gn41_20261002/` (`stats/analyze.py`); `eval3c_20261014_analysis/` (`analyze_eval3c_v2.py`, `final_out/analysis.json`); and `r2eval/` (the Rebuild-2 gates). In the October 18 run, 32 reference reranker pairs were re-scored before the run and after each of its two steps, with a maximum absolute difference of 0. The Holm adjustment of Round 3 (Section XI) is archived as output only (`verify_stats/multi.out` in the October 10 evidence snapshot `tmp_evidence_20261010/` under the store's `reports/`, with the test function in `vcommon.py`); the driver script that produced it was not archived, while `paper/tools/round3_stats.py` recomputes the unadjusted intervals.

Scale of the evaluation. A full 100-question pipeline run takes $\approx$ 2.5 hours on the single-host deployment (36 min collection, 1 h 42 m RAGAS judging, 13 min coverage judging); the retrieval-only loop takes ≈ 15 minutes and was used for the α sweep and regression forensics. On the 500-question benchmark, LLM judging alone took roughly 10–12 hours per Round 3 configuration (two faithfulness judges, three further RAGAS metrics, and coverage).

APPENDIX C
POST-AUDIT CORRECTIONS AND THE OCTOBER 2026 REBUILD

An October 2026 audit of the data pipeline contradicted several statements of earlier versions of this paper. This revision corrects them in place and leaves every experimental number of Rounds 0–3, which remain the run of record, unchanged: the corpus is the RCUV, not the new-punctuation CUV named on its PDFs, and its legacy conversion lost text and added ghost verses (Section III), so the legacy store's 31,102 verse positions are not the PDF's 31,092 (Section IV-C); the OpenBible.info TSK data are CC BY, not public domain (Section IV-C); hybrid retrieval never ran, so every round is dense-only (Section V-E); and the serving-side tokenization of Rounds 0–3 (the embedder's query encoding and the reranker's query–passage input) was inconsistent with the build side, a defect whose correction had no measurable retrieval effect (Section VI-C). The audit also measured the legacy knowledge graph of Section IV: it had no entity-identity layer — an `entity_id` was the type plus toneless pinyin, so homonyms and homophones shared one node (of the 175 edges on the apostle John's node, 29 concern the apostle) — and the strict precision of its 15,926 semantic edges, called fact edges in earlier versions, was only about 0.23–0.28 (lenient 0.34–0.45), with most wrong edges in the co-occurrence backfill (Section IV-B). The system has since been rebuilt from the 66 PDFs in versioned layers. The rebuilt text matches the PDFs with zero character-conservation residual: 31,021 verse labels (70 of them merged verses such as "29–30") over 31,092 verse positions, 11 verse numbers that the RCUV gives only in footnotes (the legacy ghost verses among them), all 116 psalm superscriptions, and 1,013 footnotes. Sparse retrieval is retired and the serving-side tokenization corrected.

Two rebuilt builds have been served: Rebuild-1 (`b20261008_6daa4f31`) and Rebuild-2 (`b20261008_e05d3e55`), the build now in service. (We avoid the labels R1/R2 for builds, since R1–R6 name routes.) Neither uses Neo4j. The knowledge layer is reduced to an L0 name layer (2,926 names), which the service uses only as routing vocabulary for person and place signals, and an L1 curated event registry (31 events in Rebuild-2; 33 in Rebuild-1, whose registry was exported from the curated events of the legacy graph); there is no entity-identity layer and no relation layer. The entity, relation, and cross-reference graph of Section IV was retired rather than rebuilt, and TSK is not loaded. The registry is a curated event lookup lane, not a graph operation: a literal trigger match appends at most one anchor passage after the completed top-k on the event and cross-book routes (R4, R5), and the appended passage takes no part in ranking. In a three-arm run on the 500 GT v2 questions (Section XII-B; one generation per arm), turning the lane on changed answer coverage by $+0.0089$ ($[-0.0030, +0.0207]$). The lane fired on only 23 of the 500 questions, and on those it adds a sixth passage, so the comparison is not k-aligned; we report it as descriptive, not as graph evidence. Rebuild-2 likewise showed no measurable retrieval gain over Rebuild-1 (verse recall@6 $+0.0002$, $[-0.0016, +0.0026]$; descriptive). In the same three-arm run, the full pipeline beat semantic-only retrieval in answer coverage by $+0.032$ ($[+0.012, +0.053]$; descriptive), a gain concentrated in the 179 questions routed to the exact-verse and chapter routes (R1, R2: $+0.092$, $[+0.048, +0.138]$), that is, in citation resolution; on the other 321 questions no gain was detected (-0.002, upper bound $+0.017$). The graph-centric serving architecture of Sections IV–V is therefore historical (Rounds 0–3); the graph-retrieval experiments on the rebuilt system are reported in Section XII.

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